RUBY

MV1-BUILD FINAL

DATE	CHANGE NO.	REV

	EE	DATE	POWER	DATE
DRAWER	MORRIS CHANG	24 / 07 / 2003		
DESIGN	MORRIS CHANG	24 / 07 / 2003		
CHECK	JOE CHANG	12 / 08 / 2003		
RESPONSIBLE	JOE CHANG	12 / 08 / 2003		

SIZE = 3 VER : A

FILE NAME : RUBY10-6050A0032201-MB-A03-13

P/N 131MBD2600A0 DATE BEGINNING : 24 / 06 / 2003

INVENTEC

TITLE

MODULES, RUBY1.0

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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8- SYSTEM POWER(1.5V/1.8V)	30- ATI-M10-P-2	51- HDD CONN
9- SYSTEM POWER(VCCP/1.2V)	31- ATI-M10-P-3	52- MULTIBAY CONN
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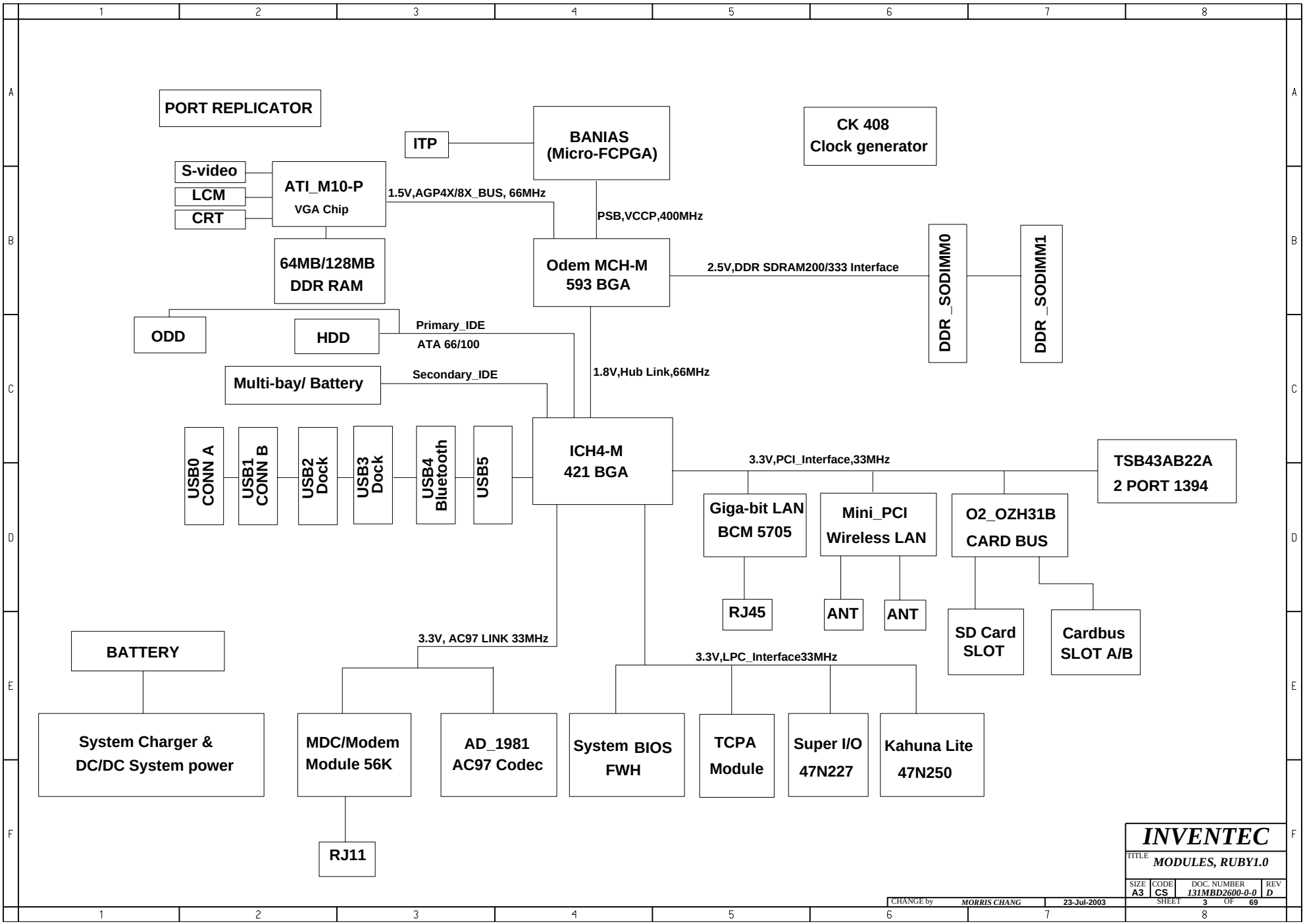
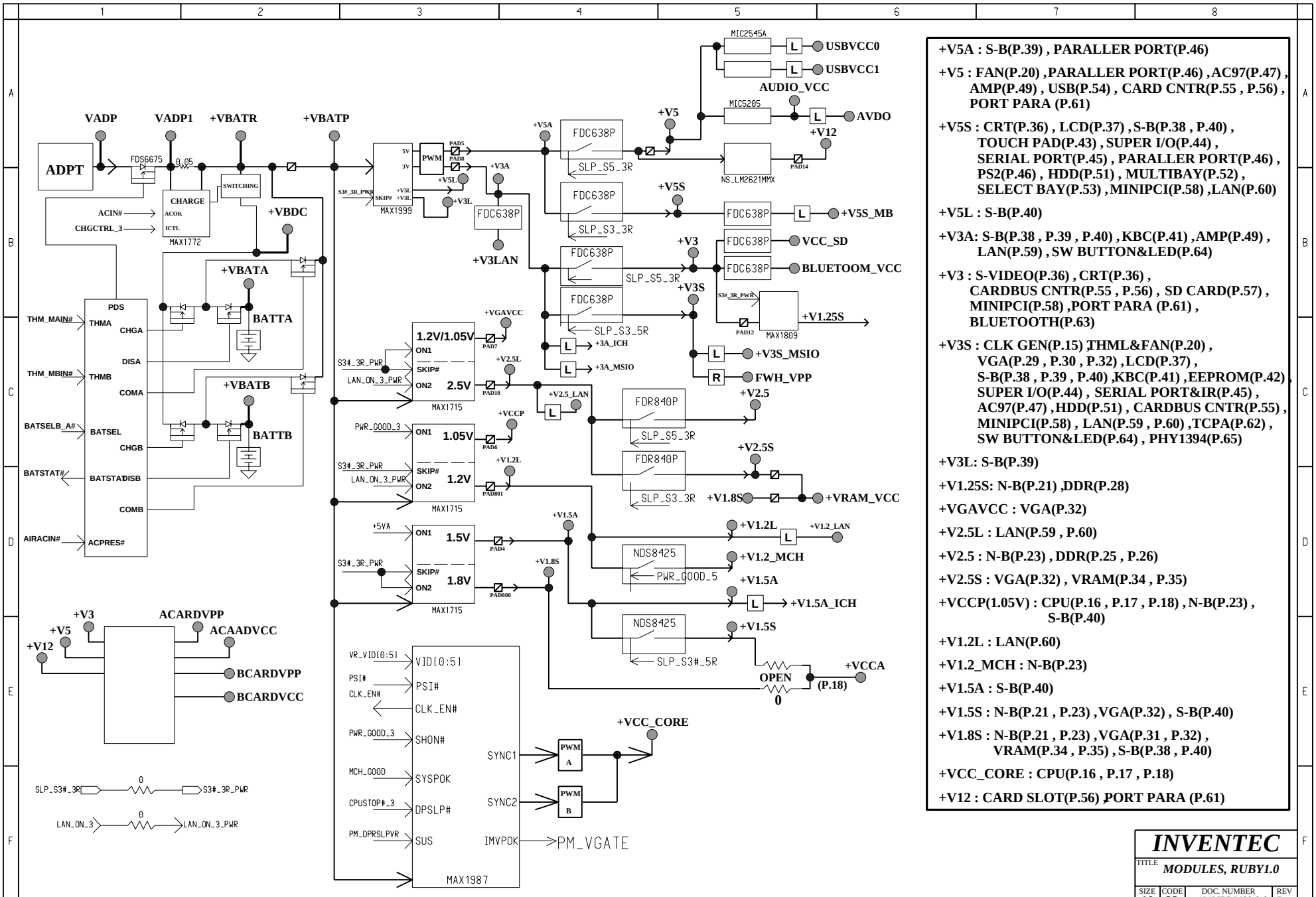
CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC			
TITLE: MODULES, RUBY1.0			
SIZE: A3	CODE: CS	DOC. NUMBER: 131MBD2600-0-0	REV: D
SHEET 2 OF 69			


PORT REPLICATOR
ITP
BANIAS
(Micro-FCPGA)
CK 408
Clock generator
S-video
LCM
CRT
ATI_M10-P
VGA Chip
1.5V,AGP4X/8X_BUS, 66MHz
PSB,VCCP,400MHz
64MB/128MB
DDR RAM
Odem MCH-M
593 BGA
2.5V,DDR SDRAM200/333 Interface
DDR_SODIMM0
DDR_SODIMM1
ODD
HDD
Primary_IDE
ATA 66/100
Multi-bay/ Battery
Secondary_IDE
1.8V,Hub Link,66MHz
USB0
CONN A
USB1
CONN B
USB2
Dock
USB3
Dock
USB4
Bluetooth
USB5
ICH4-M
421 BGA
3.3V,PCI_Interface,33MHz
TSB43AB22A
2 PORT 1394
Giga-bit LAN
BCM 5705
Mini_PCI
Wireless LAN
O2_OZH31B
CARD BUS
RJ45
ANT
ANT
SD Card
SLOT
Cardbus
SLOT A/B
BATTERY
3.3V, AC97 LINK 33MHz
3.3V,LPC_Interface33MHz
System Charger &
DC/DC System power
MDC/Modem
Module 56K
AD_1981
AC97 Codec
System BIOS
FWH
TCPA
Module
Super I/O
47N227
Kahuna Lite
47N250
RJ11
INVENTEC
TITLE
MODULES, RUBY1.0
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
CHANGE by MORRIS CHANG
23-Jul-2003
SHEET 3 OF 69



+V5A : S-B(P.39) , PARALLER PORT(P.46)

+V5 : FAN(P.20) ,PARALLER PORT(P.46) ,AC97(P.47) , AMP(P.49) , USB(P.54) , CARD CNTR(P.55 , P.56) , PORT PARA (P.61)

+V5S : CRT(P.36) , LCD(P.37) , S-B(P.38 , P.40) , TOUCH PAD(P.43) , SUPER I/O(P.44) , SERIAL PORT(P.45) , PARALLER PORT(P.46) , PS2(P.46) , HDD(P.51) , MULTIBAY(P.52) , SELECT BAY(P.53) , MINIPCI(P.58) ,LAN(P.60)

+V5L : S-B(P.40)

+V3A: S-B(P.38 , P.39 , P.40) , KBC(P.41) ,AMP(P.49) , LAN(P.59) , SW BUTTON&LED(P.64)

+V3 : S-VIDEO(P.36) , CRT(P.36) , CARDBUS CNTR(P.55 , P.56) , SD CARD(P.57) , MINIPCI(P.58) ,PORT PARA (P.61) , BLUETOOTH(P.63)

+V3S : CLK GEN(P.15) ,THML&FAN(P.20) , VGA(P.29 , P.30 , P.32) ,LCD(P.37) , S-B(P.38 , P.39 , P.40) ,KBC(P.41) ,EEPROM(P.42) , SUPER I/O(P.44) , SERIAL PORT&IR(P.45) , AC97(P.47) ,HDD(P.51) , CARDBUS CNTR(P.55) , MINIPCI(P.58) , LAN(P.59 , P.60) ,TCPA(P.62) , SW BUTTON&LED(P.64) , PHY1394(P.65)

+V3L: S-B(P.39)

+V1.25S: N-B(P.21) ,DDR(P.28)

+VGAVCC : VGA(P.32)

+V2.5L : LAN(P.59 , P.60)

+V2.5 : N-B(P.23) , DDR(P.25 , P.26)

+V2.5S : VGA(P.32) , VRAM(P.34 , P.35)

+VCCP(1.05V) : CPU(P.16 , P.17 , P.18) , N-B(P.23) , S-B(P.40)

+V1.2L : LAN(P.60)

+V1.2_MCH : N-B(P.23)

+V1.5A : S-B(P.40)

+V1.5S : N-B(P.21 , P.23) ,VGA(P.32) , S-B(P.40)

+V1.8S : N-B(P.21 , P.23) ,VGA(P.31 , P.32) , VRAM(P.34 , P.35) , S-B(P.38 , P.40)

+VCC_CORE : CPU(P.16 , P.17 , P.18)

+V12 : CARD SLOT(P.56) ,PORT PARA (P.61)

CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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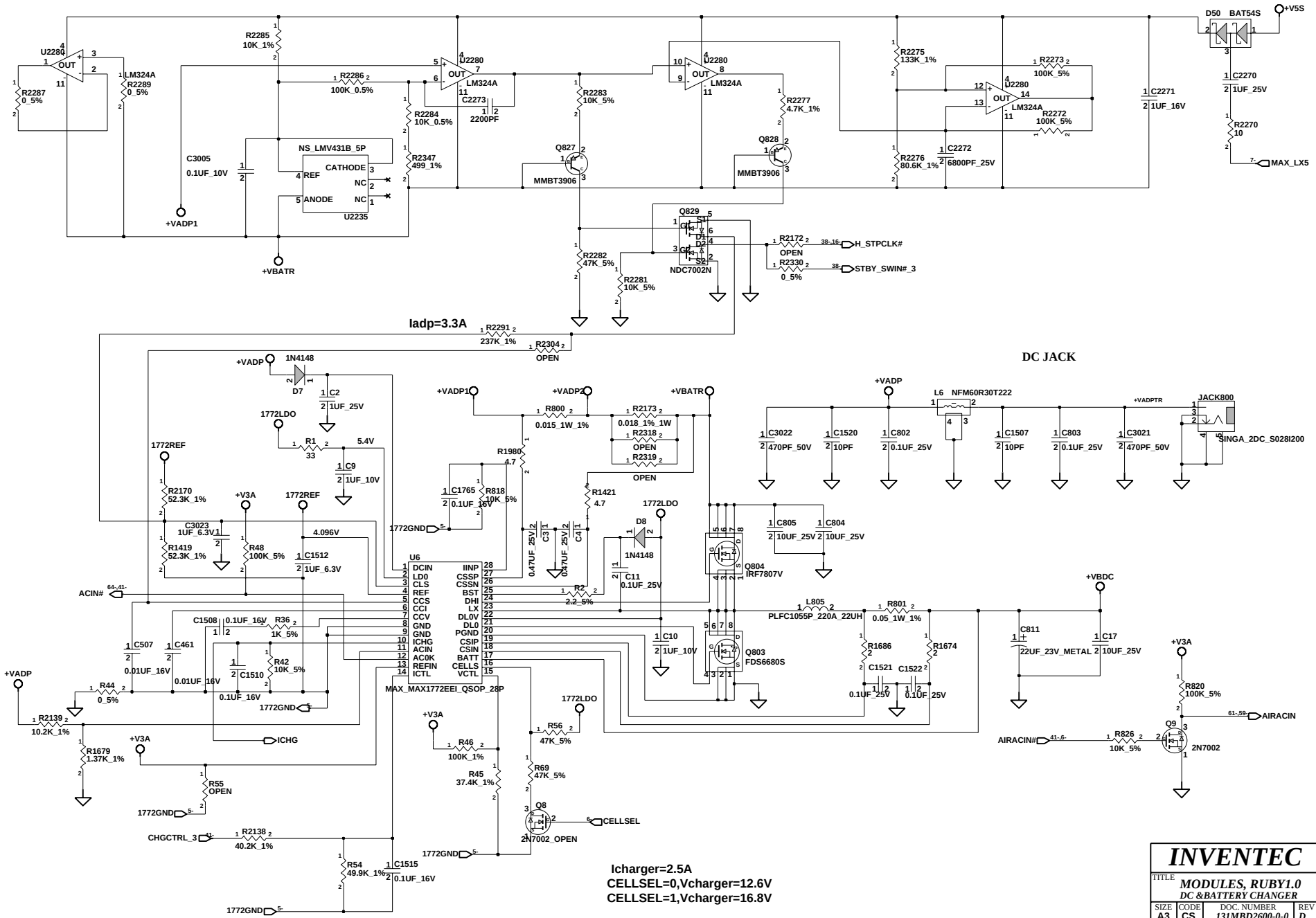
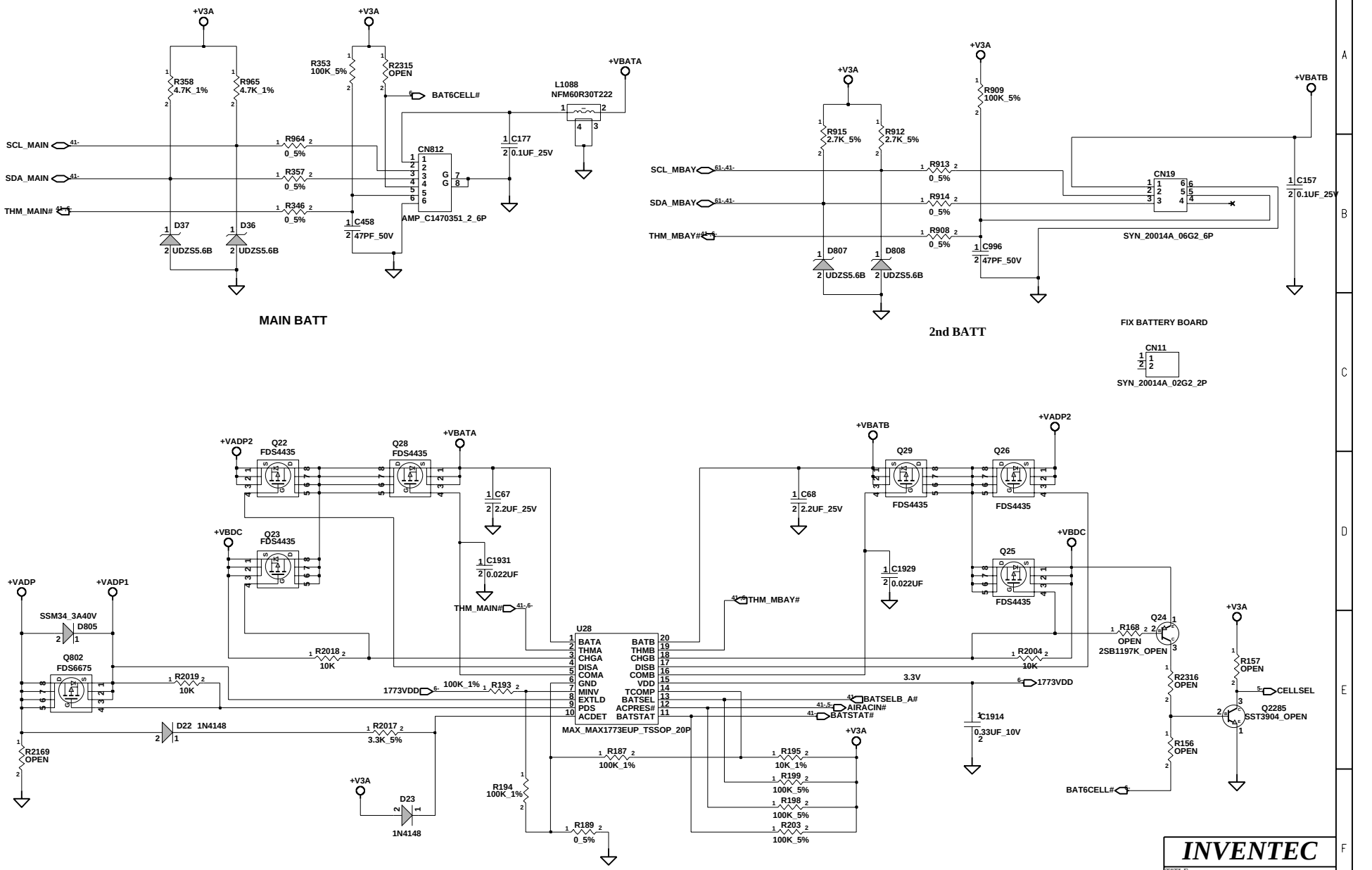
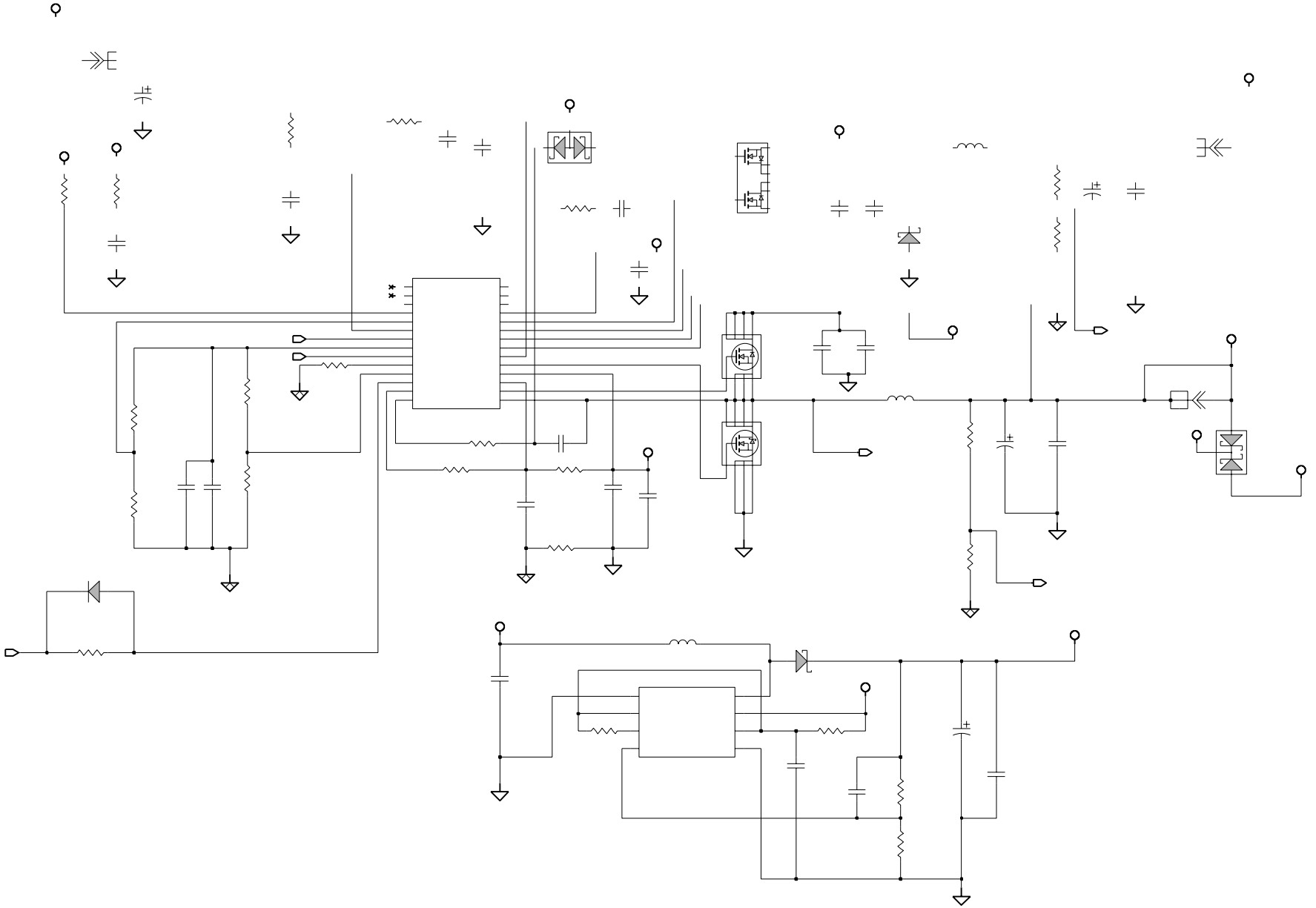
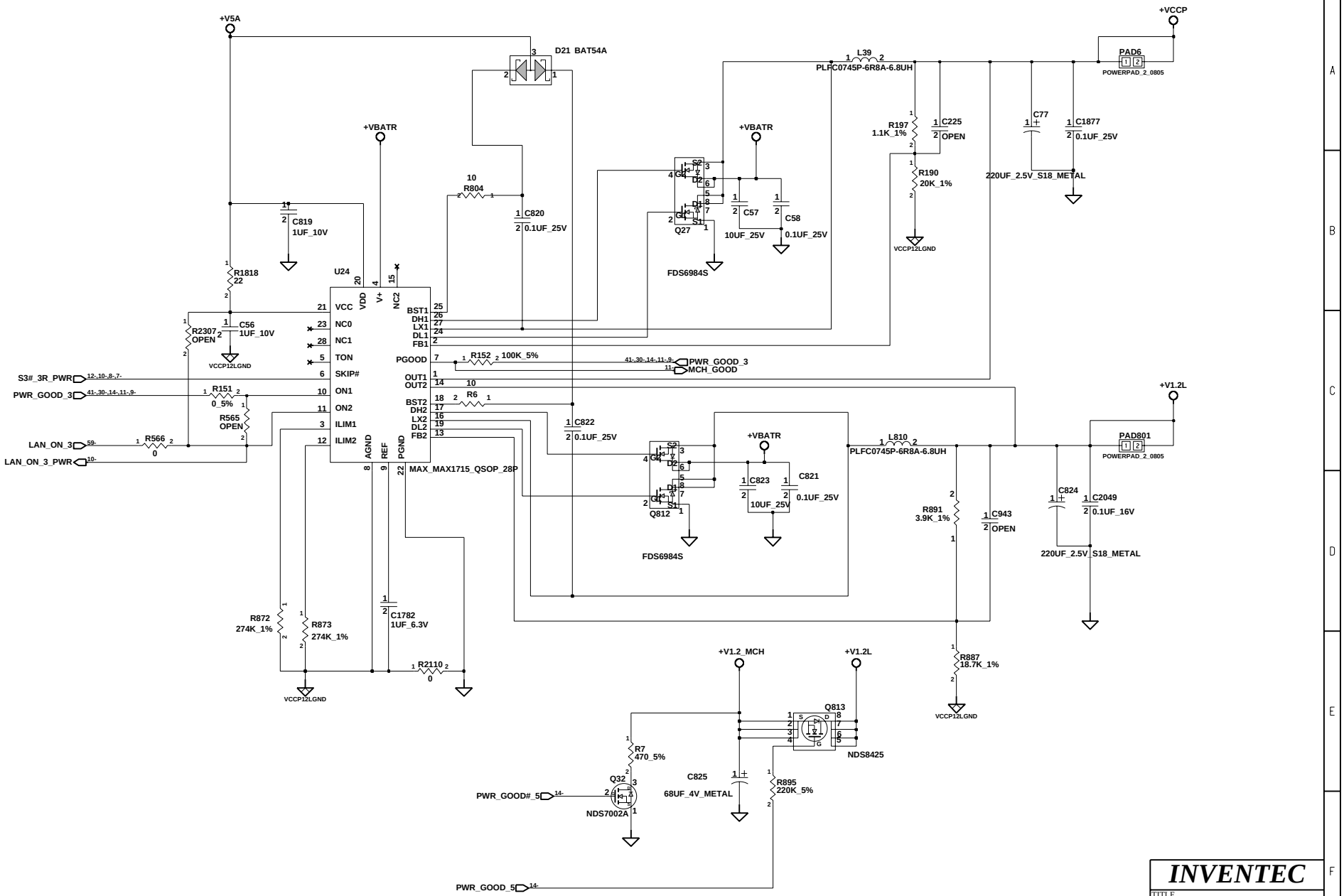

ladp=3.3A
DC JACK
Icharger=2.5A
CELLSEL=0,Vcharger=12.6V
CELLSEL=1,Vcharger=16.8V
MAX_MAX1772EEI_QSOP_28P
NS_LMV431B_5P
NDC7002N
MMBT3906
IRF7807V
FDS6680S
2N7002_OPEN
2N7002
SINGA_2DC_S028I200
PLFC1055P_220A_22UH
INVENTEC
MODULES, RUBY1.0
DC &BATTERY CHANGER
CHANGE by MORRIS CHANG
23-Jul-2003
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 5 OF 69


MAIN BATT
2nd BATT
FIX BATTERY BOARD
SYN_20014A_06G2_6P
SYN_20014A_02G2_2P
AMP_C1470351_2_6P
MAX_MAX1773EUP_TSSOP_20P
INVENTEC
MODULES, RUBY1.0
SELECT & BATTERY CONN
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 6 OF 69
CHANGE by MORRIS CHANG
23-Jul-2003

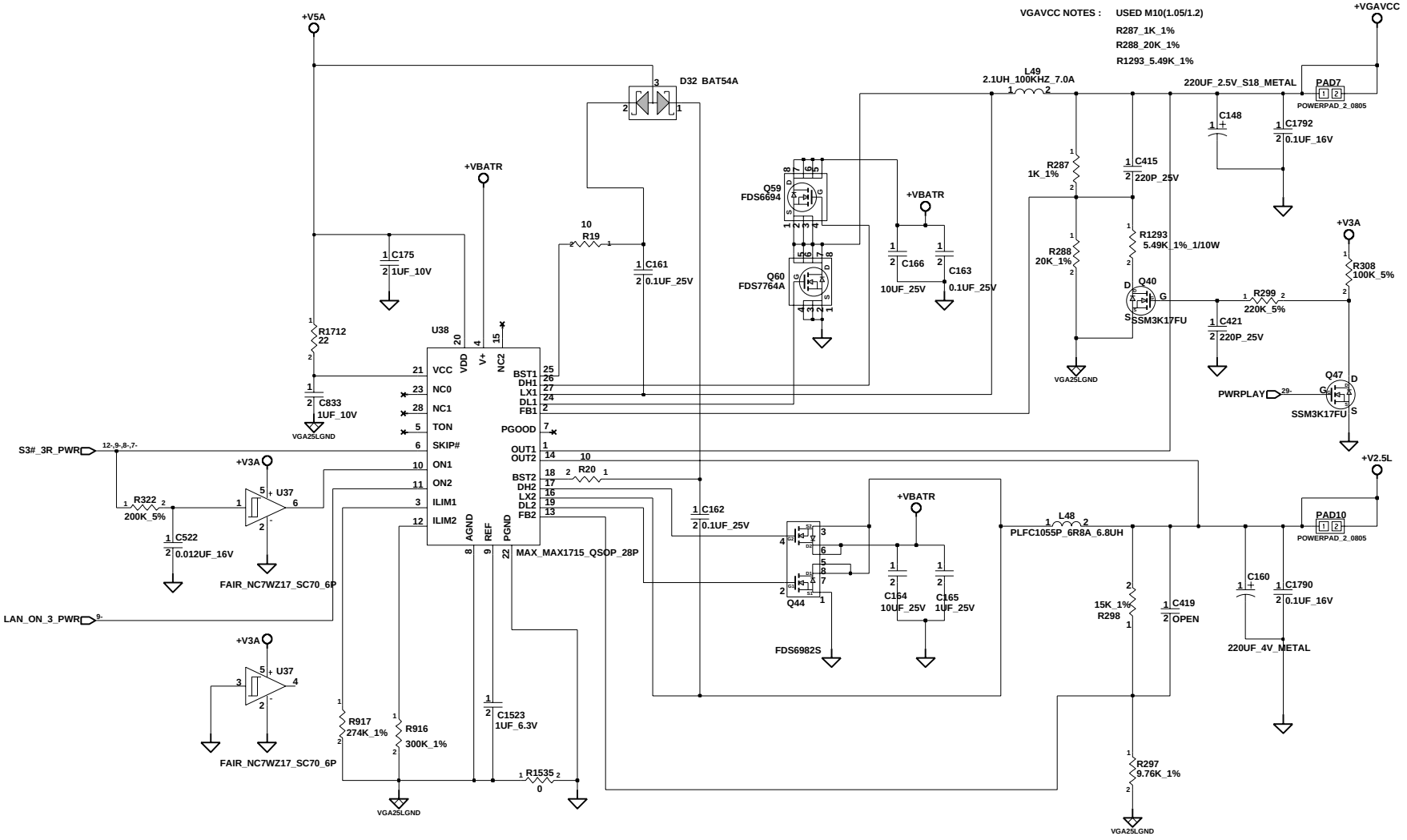




TITLE	*MODULES, RUBY1.0*
	SYSTEM POWER (VCCP/1.2V)
SIZE	A3
CODE	CS
DOC. NUMBER	*131MBD2600-0-0*
REV	*D*
SHEET	9 OF 69

CHANGE by *MORRIS CHANG* 23-Jul-2003

INVENTEC



CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
SYSTEM POWER(2.5V/1.25V)

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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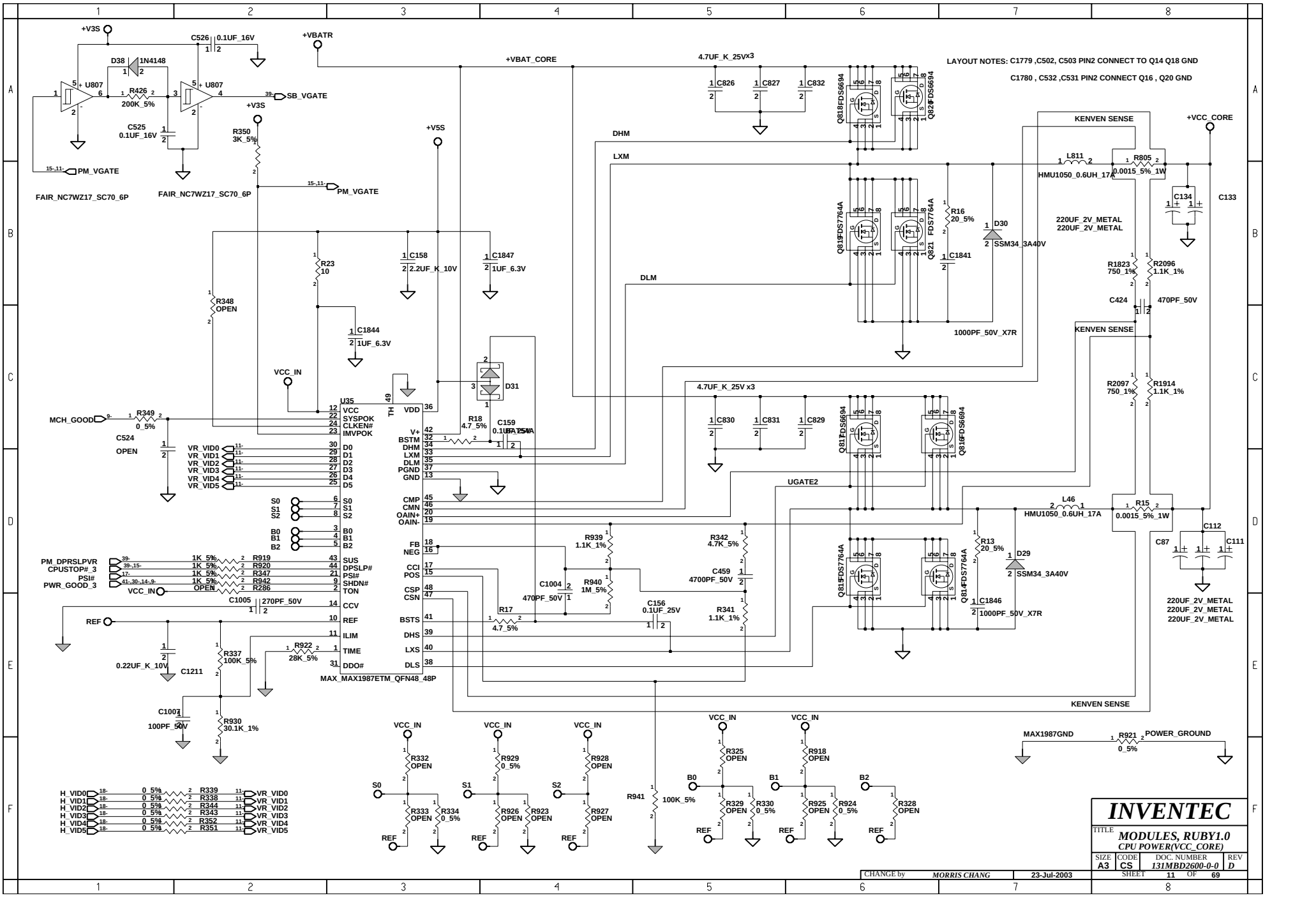
LAYOUT NOTES: C1779 ,C502, C503 PIN2 CONNECT TO Q14 Q18 GND
C1780 , C532 ,C531 PIN2 CONNECT Q16 , Q20 GND
+V3S
+VBATR
+VBAT_CORE
+V5S
+VCC_CORE
VCC_IN
C526 0.1UF_16V
D38 1N4148
U807
R426 200K_5%
SB_VGATE
C525 0.1UF_16V
R350 3K_5%
PM_VGATE
FAIR_NC7WZ17_SC70_6P
4.7UF_K_25V x3
C826
C827
C832
Q818 FDS6694
Q824 FDS6694
DHM
LXM
DLM
KENVEN SENSE
L811 HMU1050_0.6UH_17A
R805 0.0015_5%_1W
C133
220UF_2V_METAL
220UF_2V_METAL
C134
Q819 FDS7764A
Q821 FDS7764A
R16 20_5%
D30 SSM34_3A40V
C1841
1000PF_50V_X7R
R1823 750_1%
R2096 1.1K_1%
C424 470PF_50V
R2097 750_1%
R1914 1.1K_1%
R23 10
C158 2.2UF_K_10V
C1847 1UF_6.3V
R348 OPEN
C1844 1UF_6.3V
D31
U35
MAX_MAX1987ETM_QFN48_48P
VCC
SYSPOK
CLKEN#
IMVPOK
TH
VDD
V+
BSTM
DHM
LXM
DLM
PGND
GND
CMP
CMN
OAIN+
OAIN-
FB
NEG
CCI
POS
CSP
CSN
CCV
REF
ILIM
TIME
DDO#
BSTS
DHS
LXS
DLS
D0
D1
D2
D3
D4
D5
S0
S1
S2
B0
B1
B2
SUS
DPSLP#
PSI#
SHDN#
TON
MCH_GOOD
R349 0_5%
C524 OPEN
VR_VID0
VR_VID1
VR_VID2
VR_VID3
VR_VID4
VR_VID5
R18 4.7_5%
C159 0.1UF_25V
UGATE2
C830
C831
C829
Q817 FDS6694
Q823 FDS6694
L46 HMU1050_0.6UH_17A
R15 0.0015_5%_1W
C112
C87
C111
220UF_2V_METAL
220UF_2V_METAL
220UF_2V_METAL
R939 1.1K_1%
R940 1M_5%
C1004 470PF_50V
R342 4.7K_5%
C459 4700PF_50V
R341 1.1K_1%
C156 0.1UF_25V
R17 4.7_5%
Q815 FDS7764A
Q814 FDS7764A
R13 20_5%
D29 SSM34_3A40V
C1846 1000PF_50V_X7R
PM_DPRSLPVR
CPUSTOP#_3
PSI#
PWR_GOOD_3
R919 1K_5%
R920 1K_5%
R947 1K_5%
R942 1K_5%
R286 OPEN
C1005 270PF_50V
R922 28K_5%
C1211 0.22UF_K_10V
R337 100K_5%
C1007 100PF_50V
R930 30.1K_1%
R332 OPEN
R333 OPEN
R334 0_5%
R929 0_5%
R926 OPEN
R923 OPEN
R928 OPEN
R927 OPEN
R941 100K_5%
R325 OPEN
R329 OPEN
R330 0_5%
R918 OPEN
R925 OPEN
R924 0_5%
R328 OPEN
MAX1987GND
R921 0_5%
POWER_GROUND
H_VID0
H_VID1
H_VID2
H_VID3
H_VID4
H_VID5
R339
R338
R344
R343
R352
R351
INVENTEC
MODULES, RUBY1.0
CPU POWER(VCC_CORE)
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
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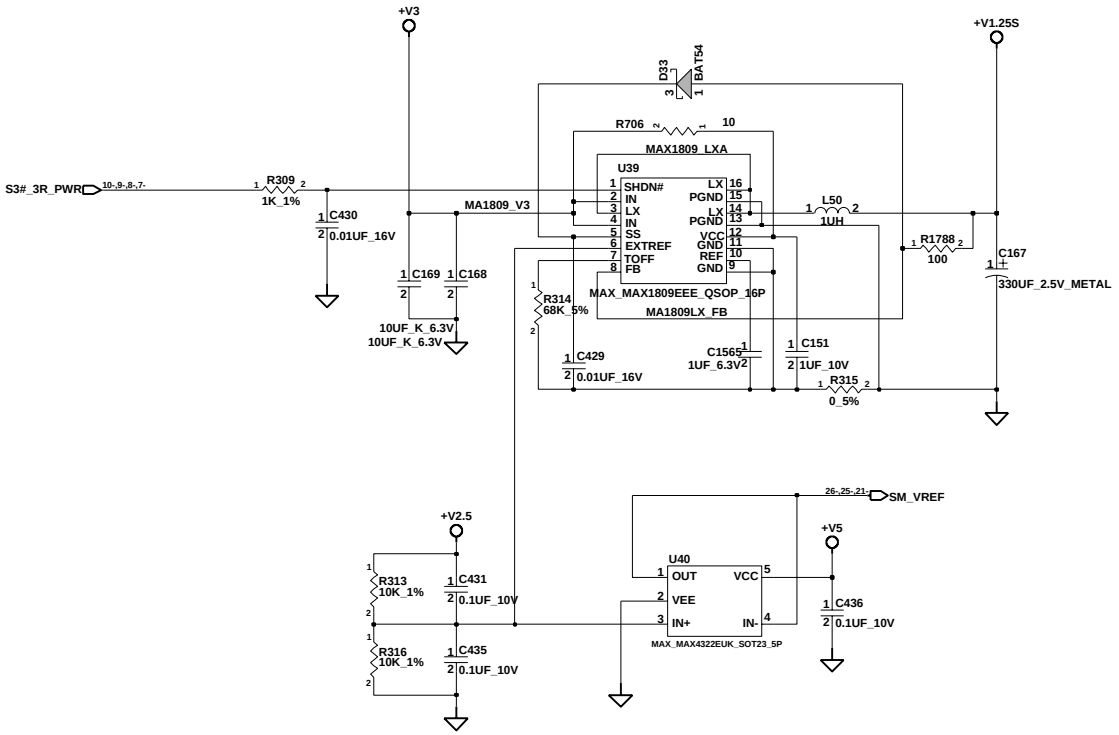
CHANGE by MORRIS CHANG
23-Jul-2003



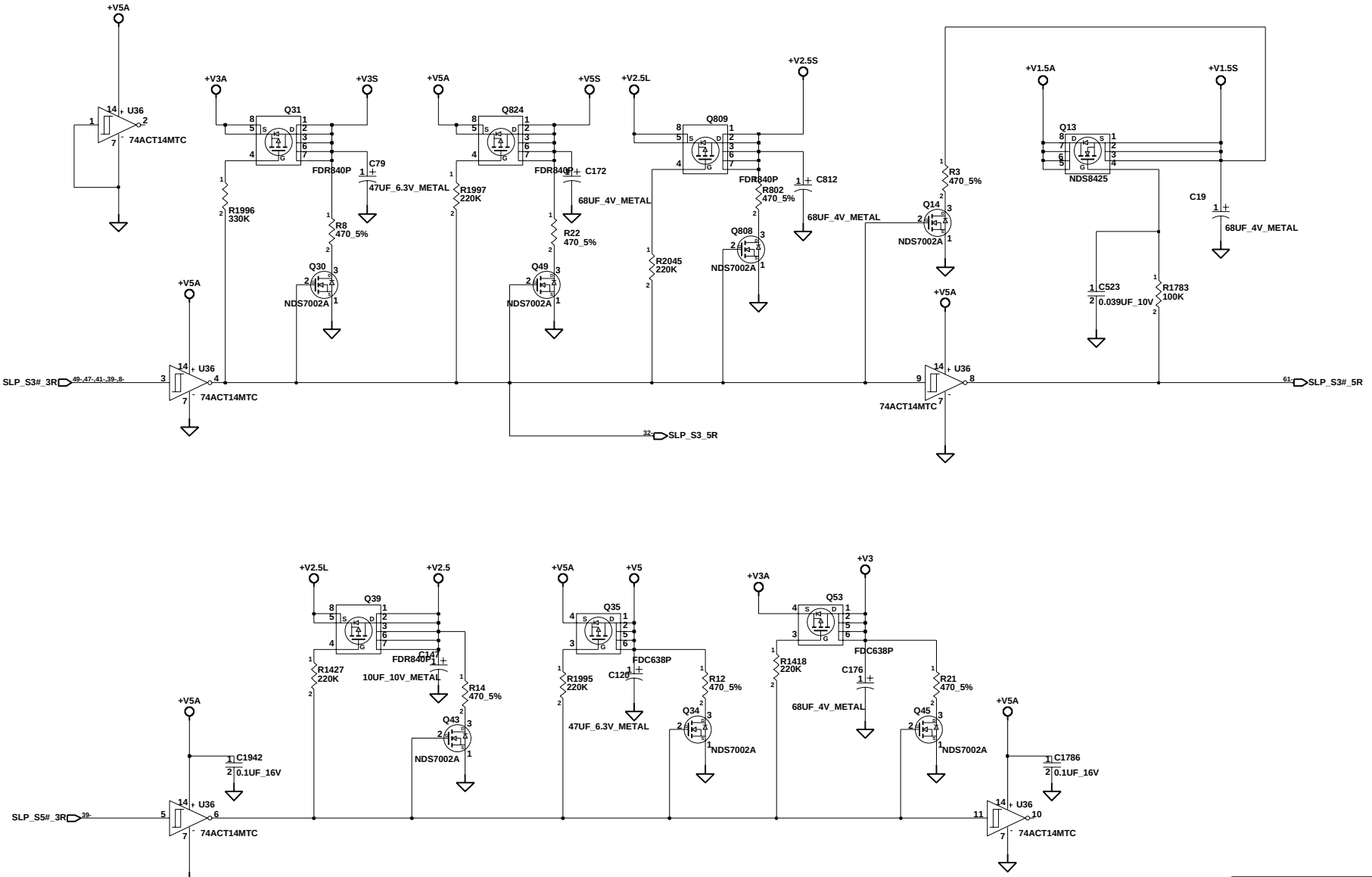
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

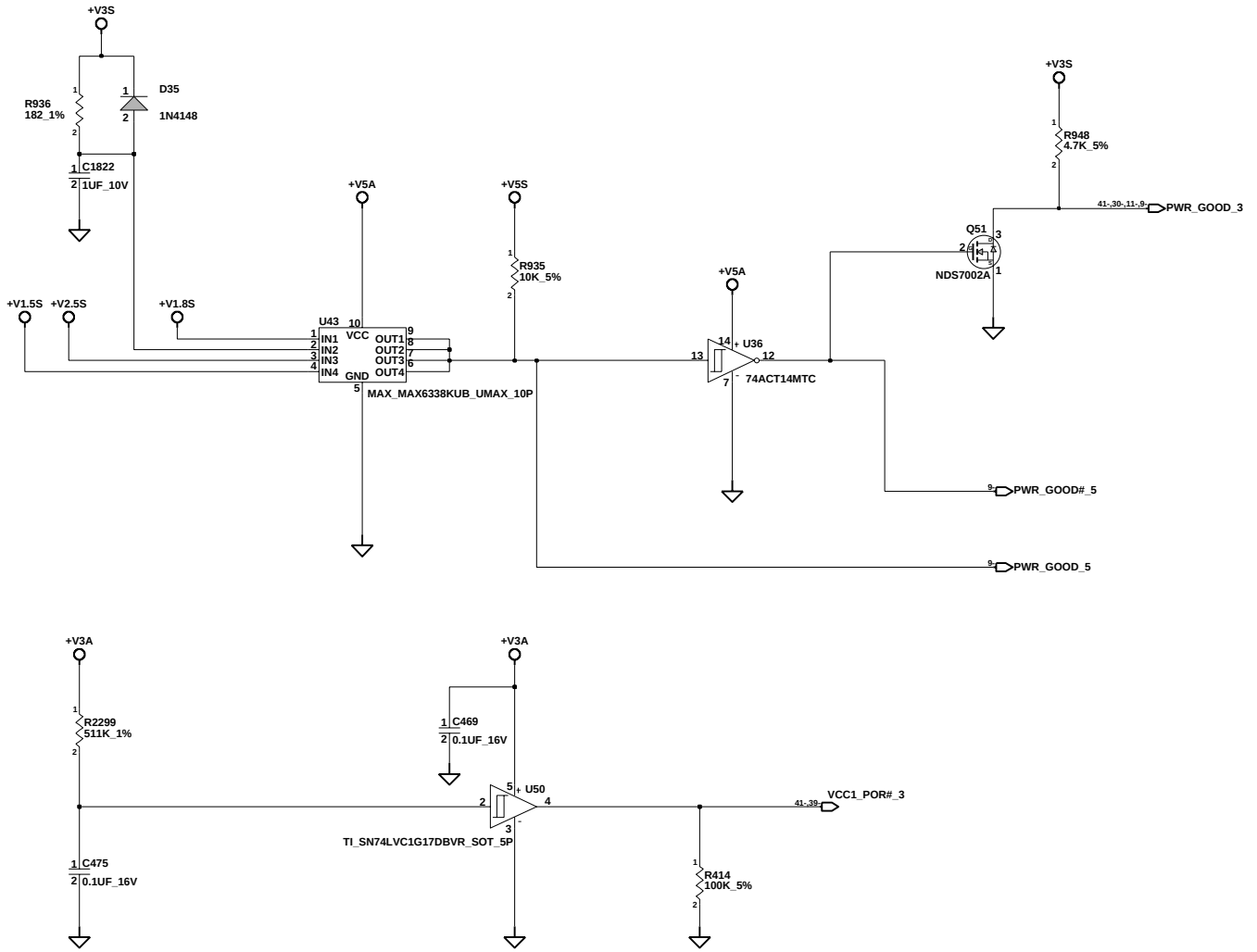
TITLE: ***MODULES, RUBY1.0***
DDR 1.25V POWER

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 12 OF 69



INVENTEC			
TITLE	*MODULES, RUBY1.0* *POWER(SLEEP)*		
SIZE **A3**	CODE **CS**	DOC. NUMBER *131MBD2600-0-0*	REV *D*
CHANGE by *MORRIS CHANG*	**23-Jul-2003**	SHEET **13** OF **69**	



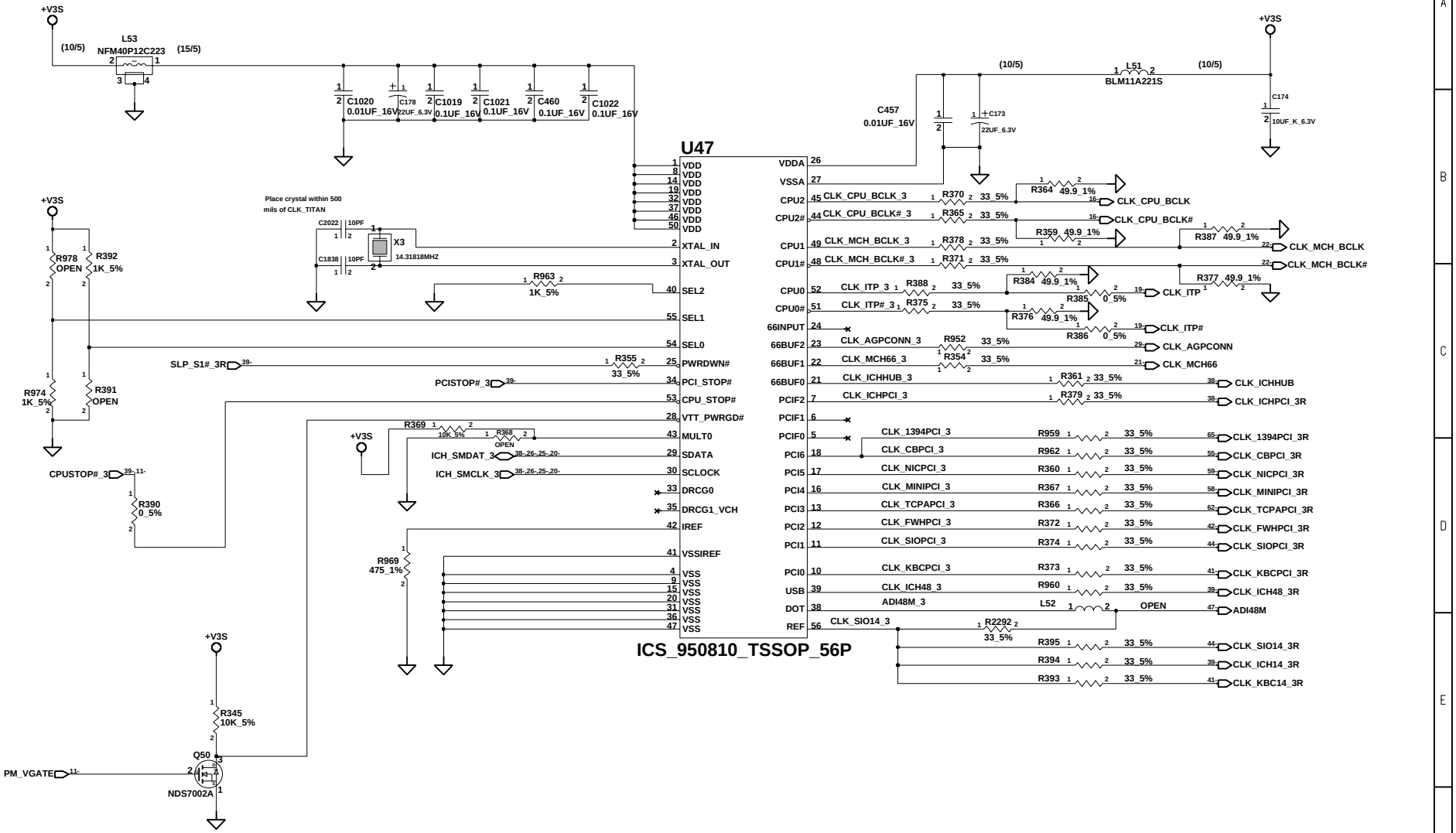
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
POWER(SEQUENCE)

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 14 OF 69



CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
CLOCK_GENEATOR

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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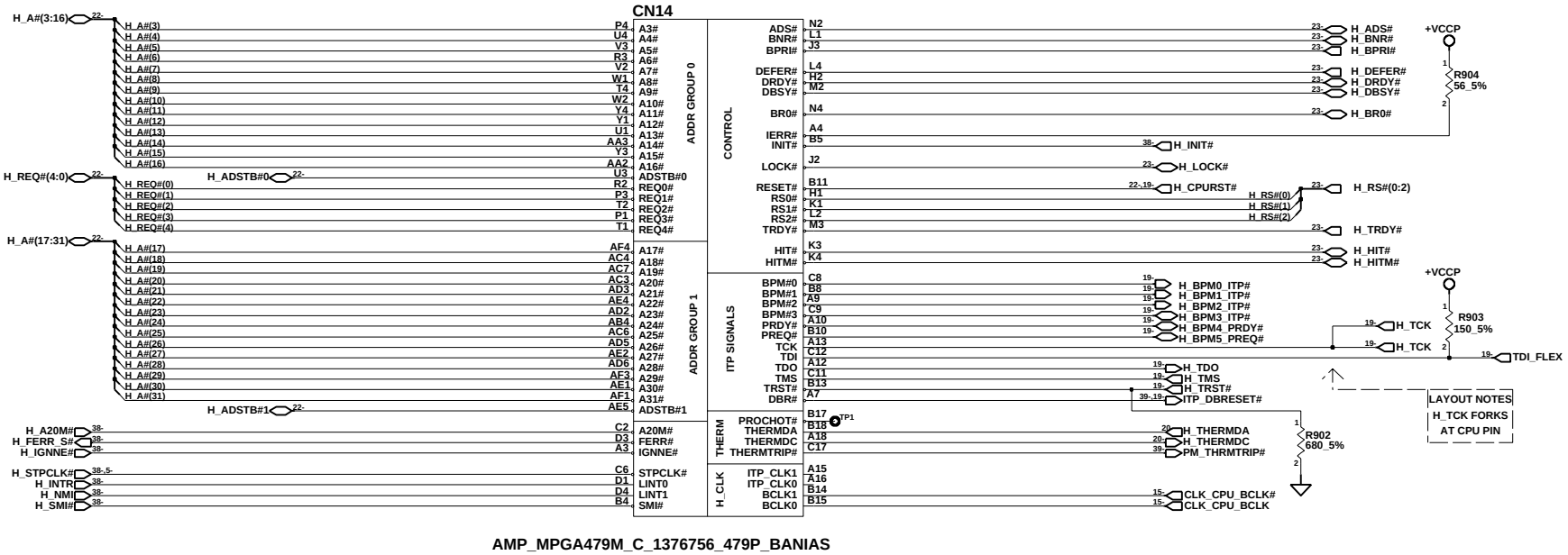
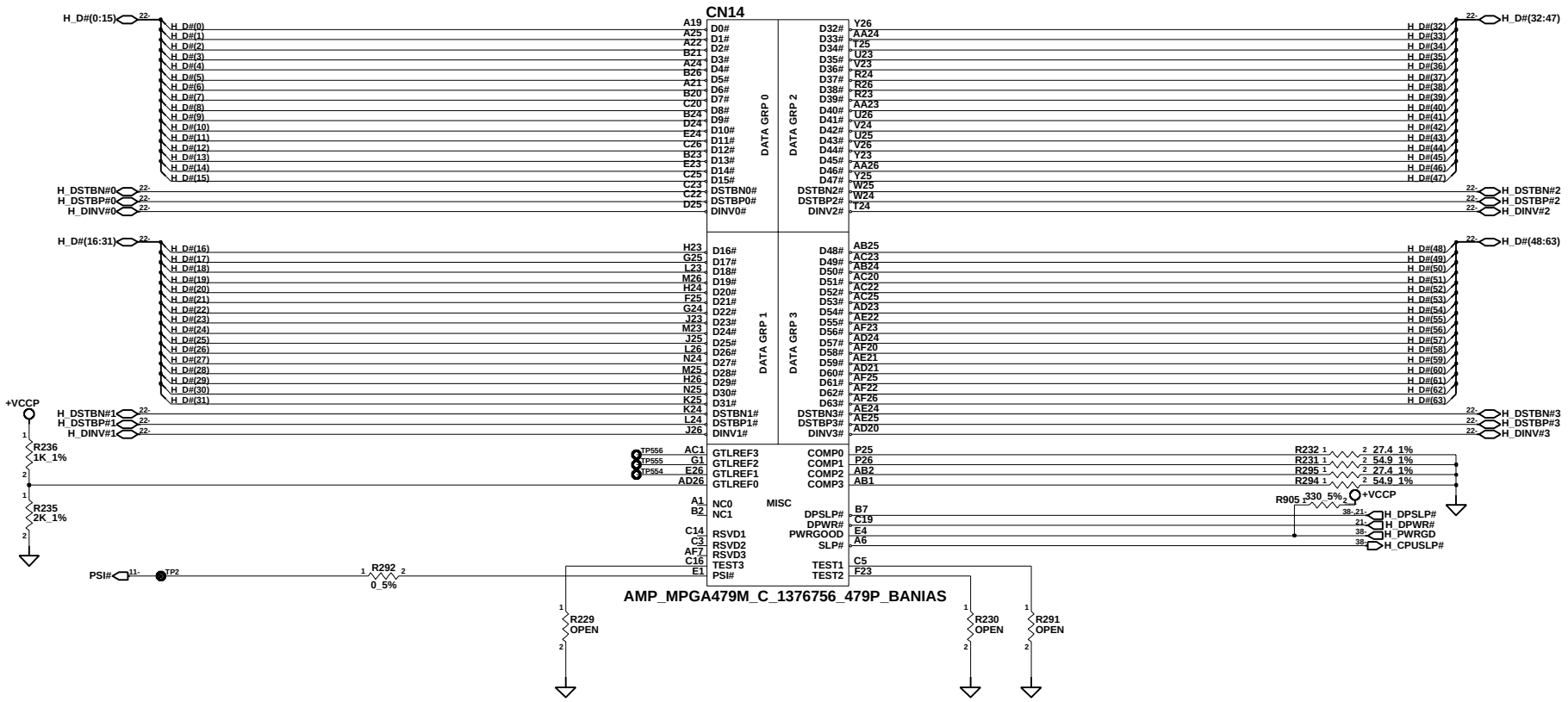


AMP_MPGA479M_C_1376756_479P_BANIAS

CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC				
TITLE	MODULES, RUBY1.0 BANIAS-1			
SIZE	CODE	DOC. NUMBER		REV
A3	CS	131MBD2600-0-0		D
		SHEET 16	OF 69	



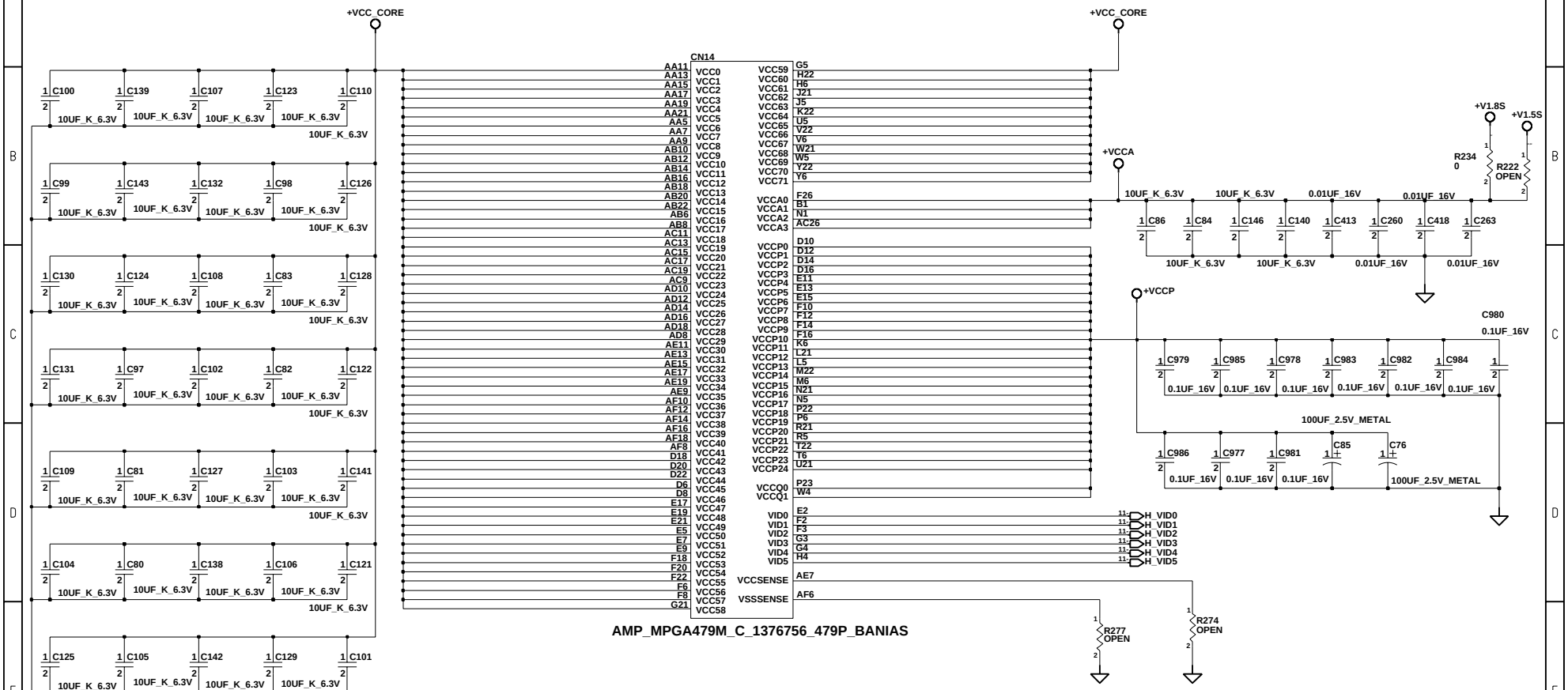
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
BANIAS-2

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 17 OF 69



CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
BANIAS-3

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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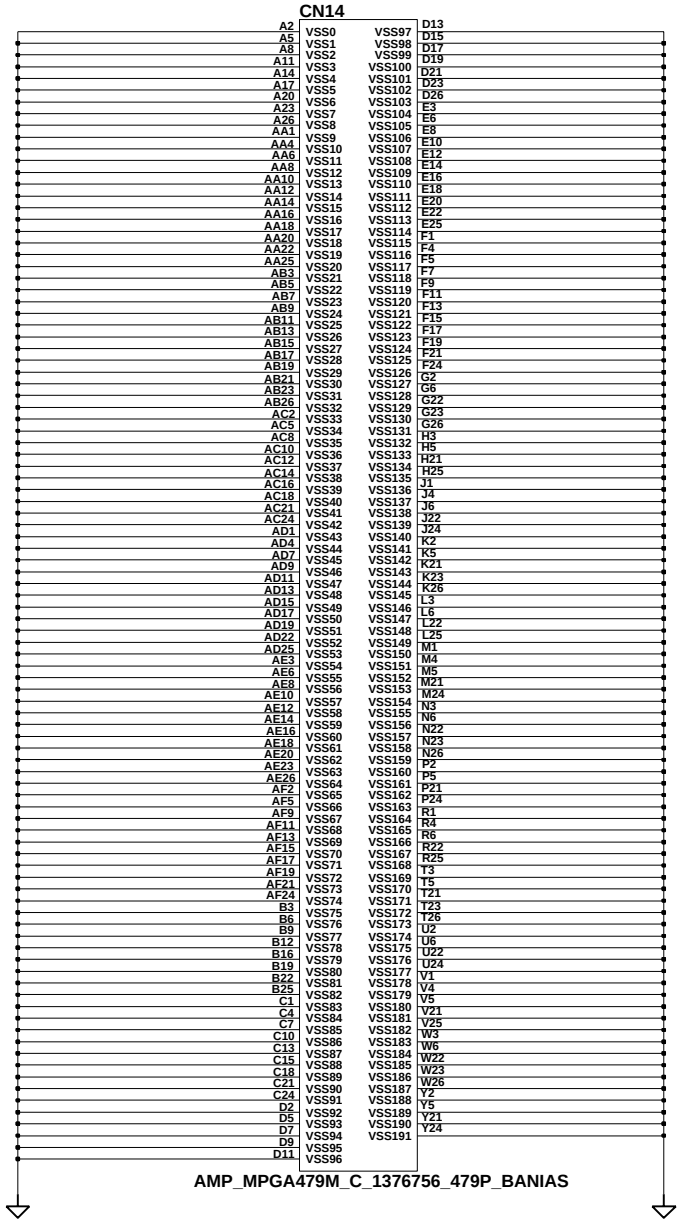
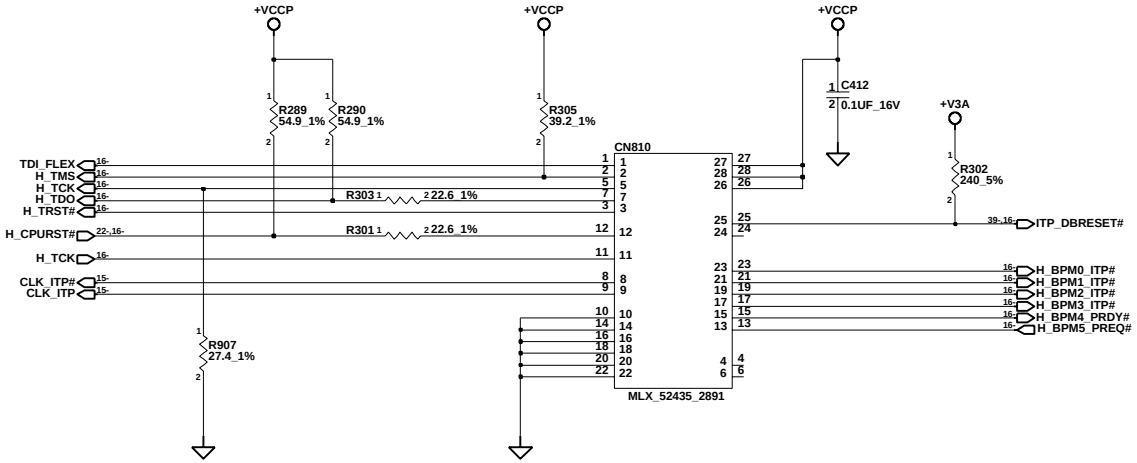


In-Target Probe



LAYOUT NOTES :

H_CPURST# PIN OF MCH FORK OUT INTO TWO BRANCH ON CPU AND ITP CONNECTOR

CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
BANIAS-4

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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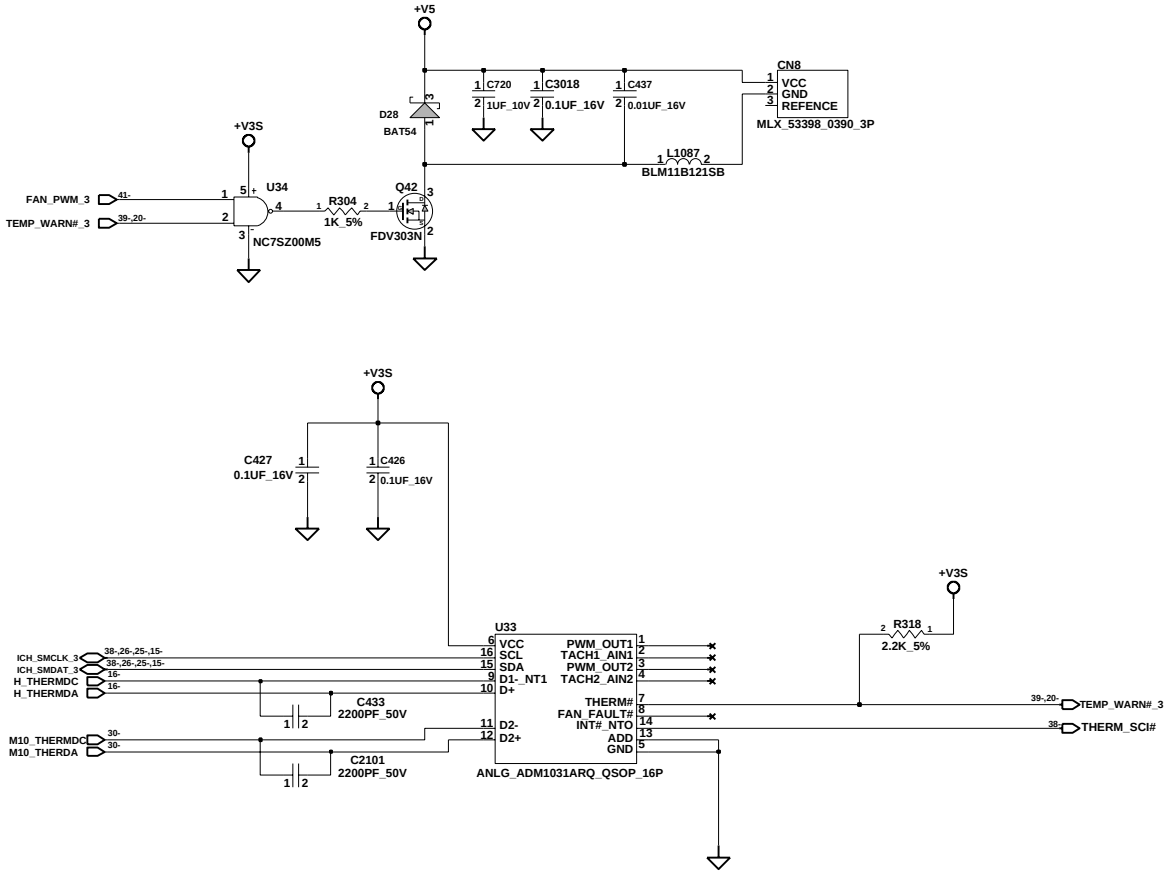

LAYOUT NOTES : PUT THE THERMAL SENSOR CLOSE TO CPU

CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
THERMAL&FAN CONTROLLER

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

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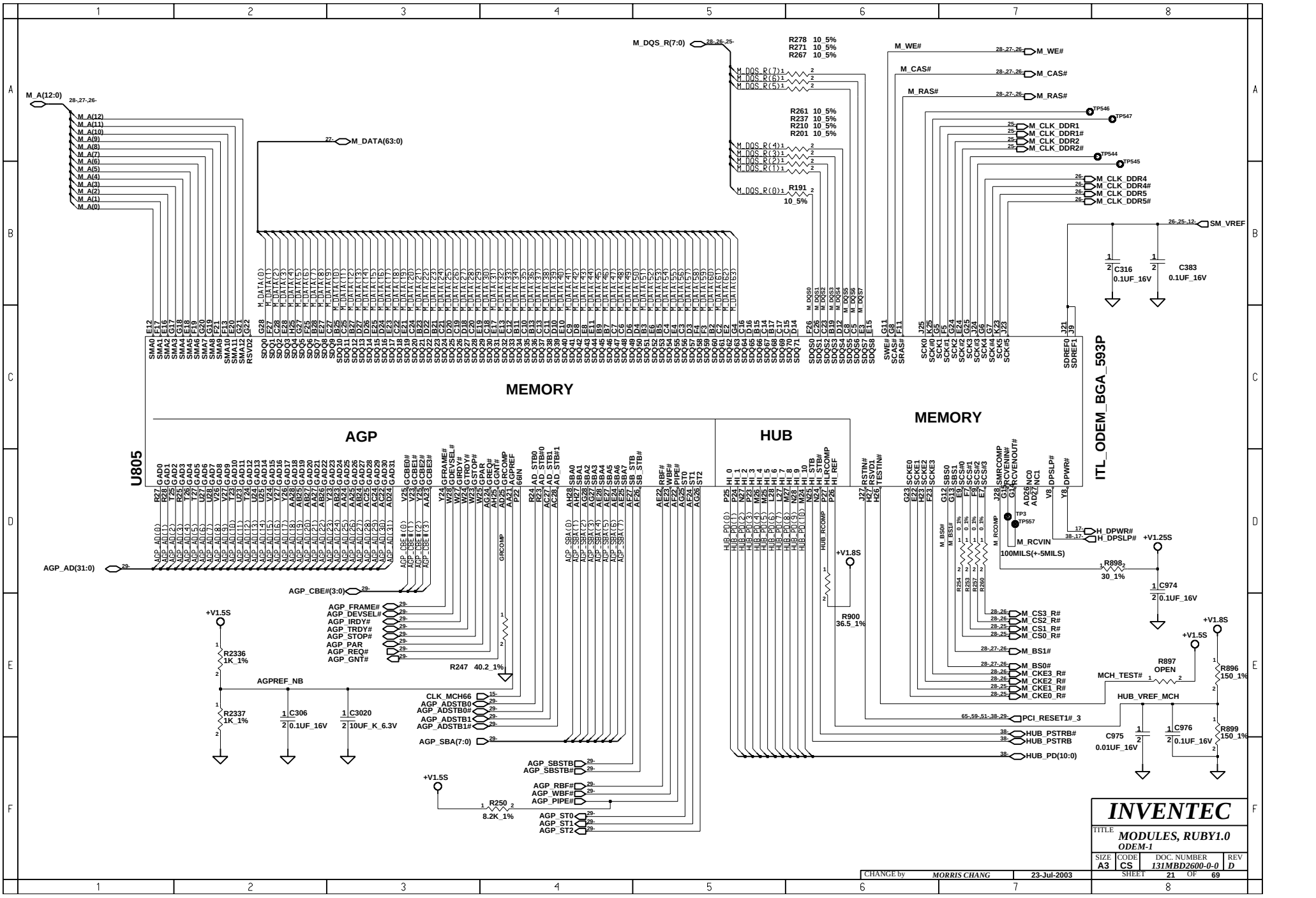
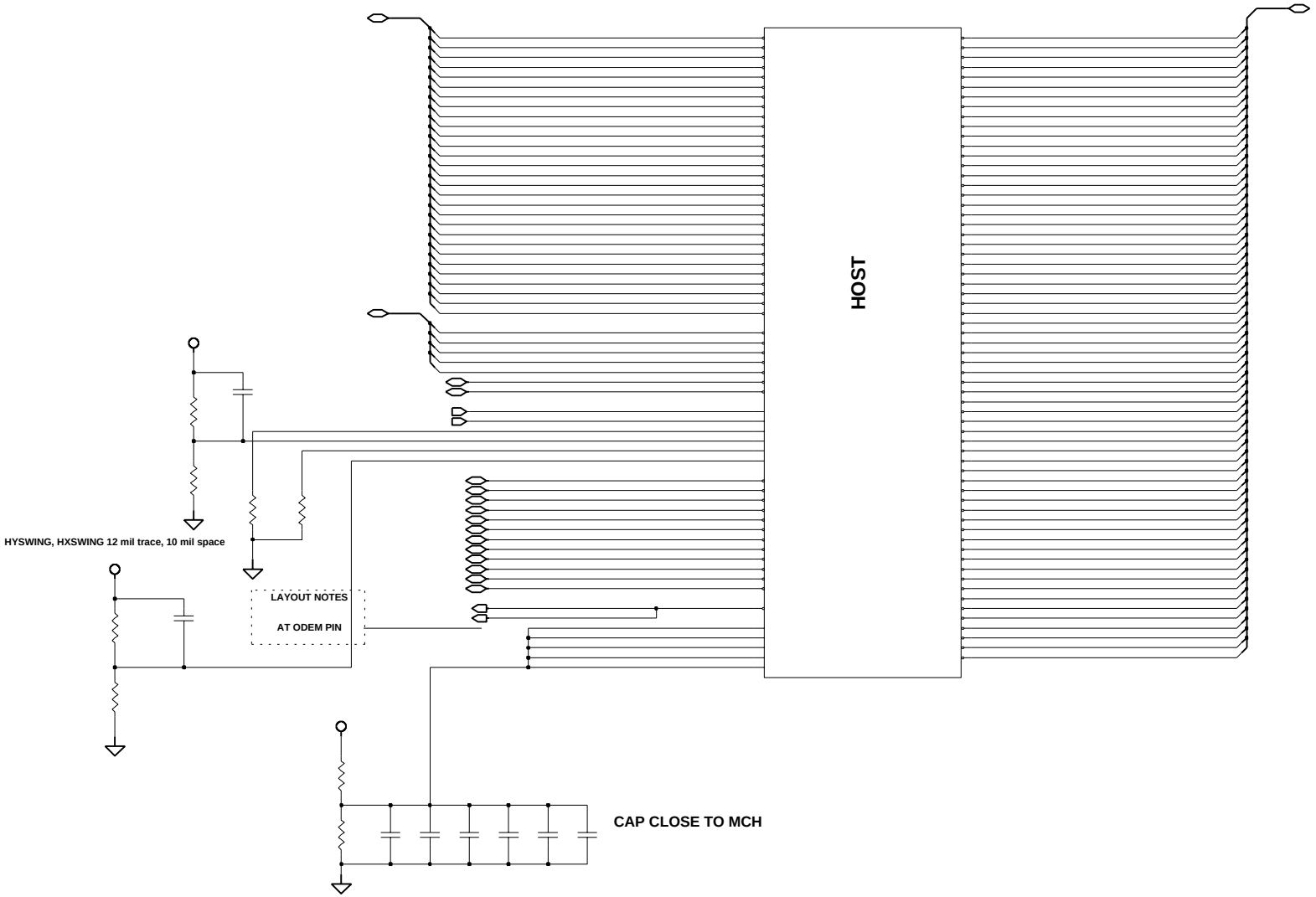
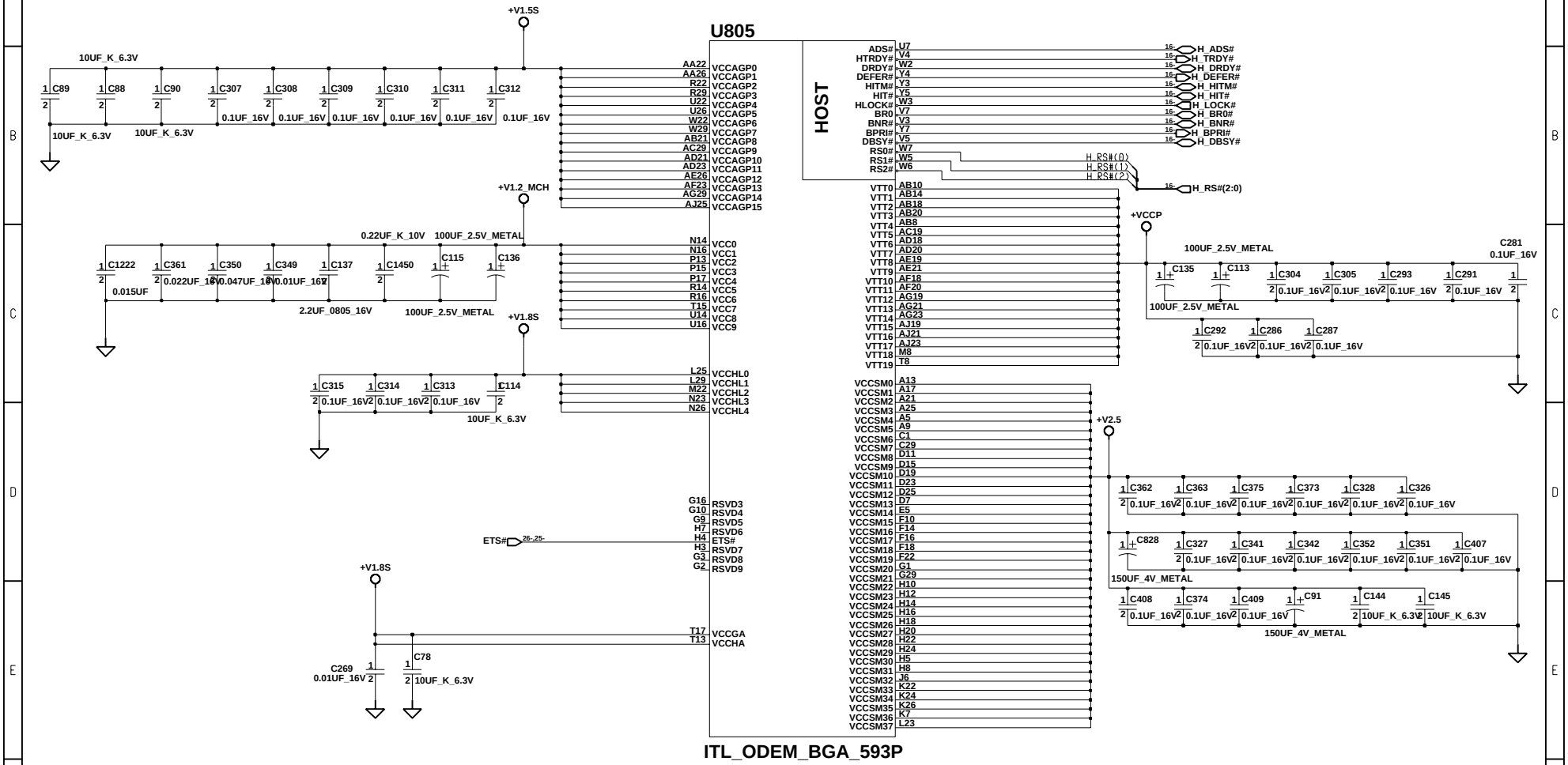

M_A(12:0)
M_DATA(63:0)
M_DQS_R(7:0)
R278 10_5%
R271 10_5%
R267 10_5%
R261 10_5%
R237 10_5%
R210 10_5%
R201 10_5%
R191 10_5%
M_WE#
M_CAS#
M_RAS#
M_CLK_DDR1
M_CLK_DDR1#
M_CLK_DDR2
M_CLK_DDR2#
M_CLK_DDR4
M_CLK_DDR4#
M_CLK_DDR5
M_CLK_DDR5#
SM_VREF
C316 0.1UF_16V
C383 0.1UF_16V
MEMORY
MEMORY
ITL_ODEM_BGA_593P
U805
AGP
HUB
AGP_AD(31:0)
AGP_CBE#(3:0)
AGP_FRAME#
AGP_DEVSEL#
AGP_IRDY#
AGP_TRDY#
AGP_STOP#
AGP_PAR
AGP_REQ#
AGP_GNT#
R247 40.2_1%
AGPREF_NB
R2336 1K_1%
R2337 1K_1%
C306 0.1UF_16V
C3020 10UF_K_6.3V
+V1.5S
+V1.8S
+V1.25S
R900 36.5_1%
R898 30_1%
C974 0.1UF_16V
H_DPWR#
H_DPSLP#
M_RCVIN
100MILS(+-5MILS)
M_CS3_R#
M_CS2_R#
M_CS1_R#
M_CS0_R#
M_BS1#
M_BS0#
M_CKE3_R#
M_CKE2_R#
M_CKE1_R#
M_CKE0_R#
R897 OPEN
MCH_TEST#
R896 150_1%
HUB_VREF_MCH
PCI_RESET1#_3
HUB_PSTRB#
HUB_PSTRB
HUB_PD(10:0)
C975 0.01UF_16V
C976 0.1UF_16V
R899 150_1%
CLK_MCH66
AGP_ADSTB0
AGP_ADSTB0#
AGP_ADSTB1
AGP_ADSTB1#
AGP_SBA(7:0)
AGP_SBSTB
AGP_SBSTB#
AGP_RBF#
AGP_WBF#
AGP_PIPE#
AGP_ST0
AGP_ST1
AGP_ST2
R250 8.2K_1%
INVENTEC
MODULES, RUBY1.0
ODEM-1
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 21 OF 69
CHANGE by MORRIS CHANG
23-Jul-2003

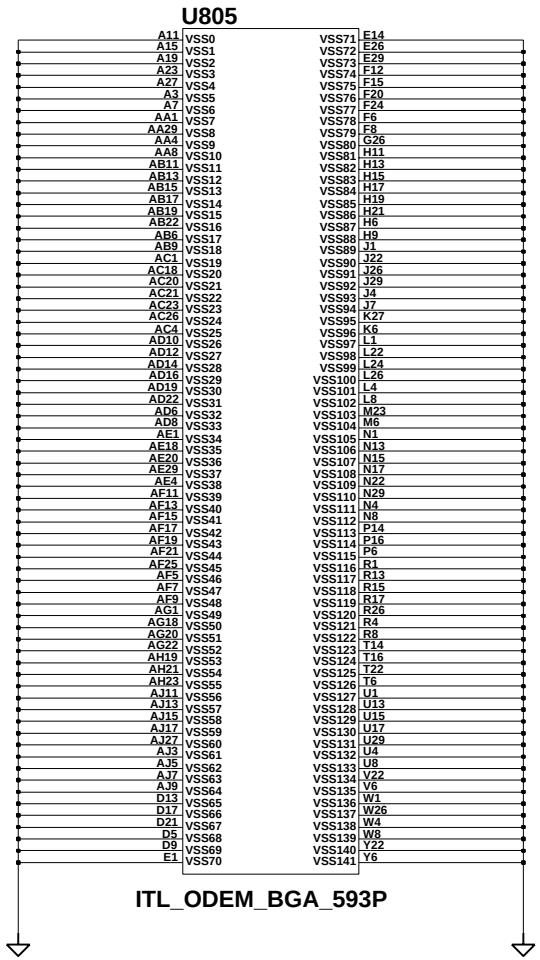


MORRIS CHANG	23-Jul-2003		D



TITLE	*MODULES, RUBY1.0* *ODEM-3*
SIZE	A3
CODE	CS
DOC. NUMBER	*131MBD2600-0-0*
REV	*D*
SHEET	23 OF 69
CHANGE by	*MORRIS CHANG*
	23-Jul-2003

INVENTEC



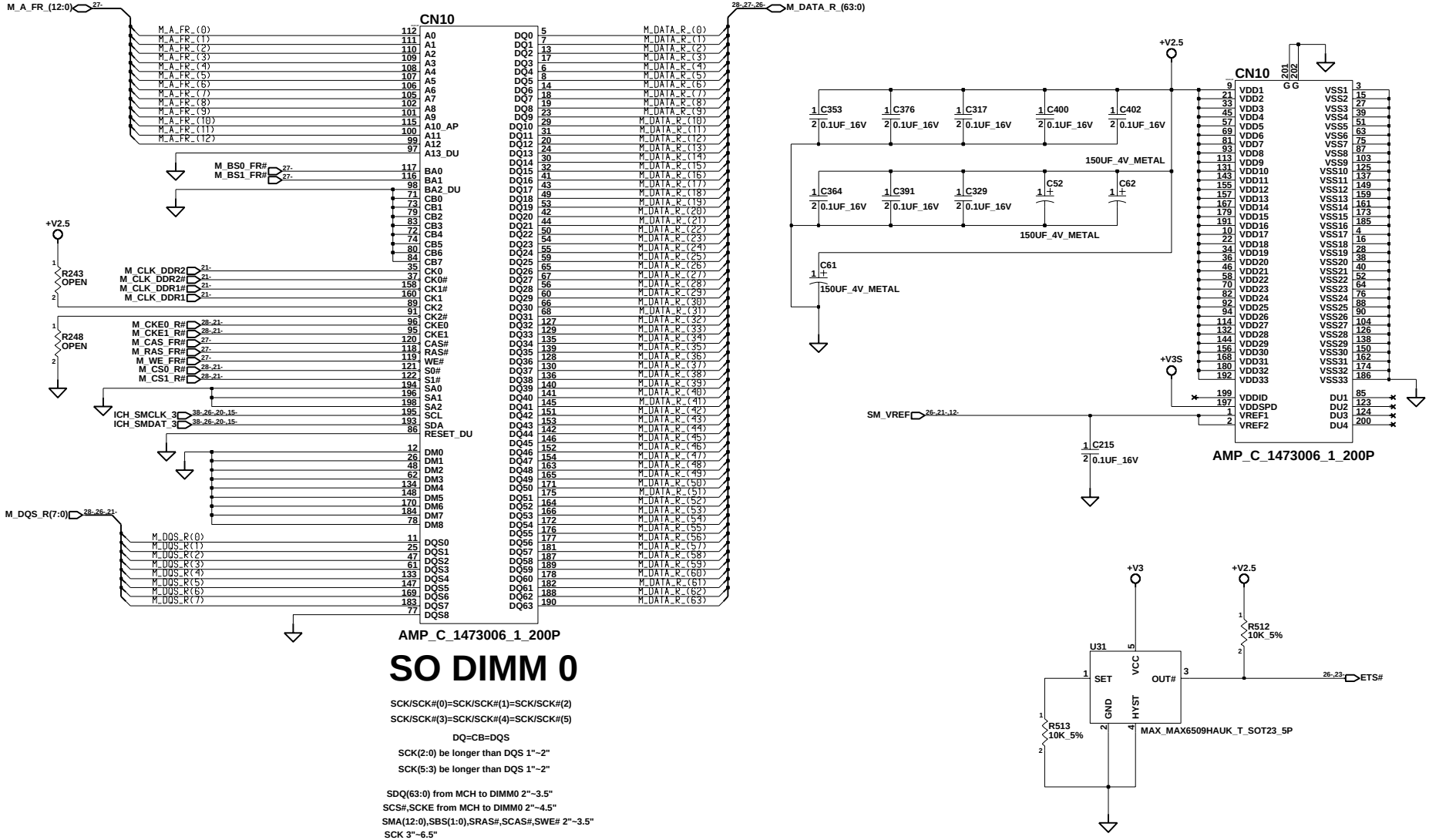
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
ODEM-4

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 24 OF 69



SO DIMM 0

SCK/SCK#(0)=SCK/SCK#(1)=SCK/SCK#(2)

SCK/SCK#(3)=SCK/SCK#(4)=SCK/SCK#(5)

DQ=CB=DQS

SCK(2:0) be longer than DQS 1"~2"

SCK(5:3) be longer than DQS 1"~2"

SDQ(63:0) from MCH to DIMM0 2"~3.5"

SCS#,SCKE from MCH to DIMM0 2"~4.5"

SMA(12:0),SBS(1:0),SRAS#,SCAS#,SWE# 2"~3.5"

SCK 3"~6.5"

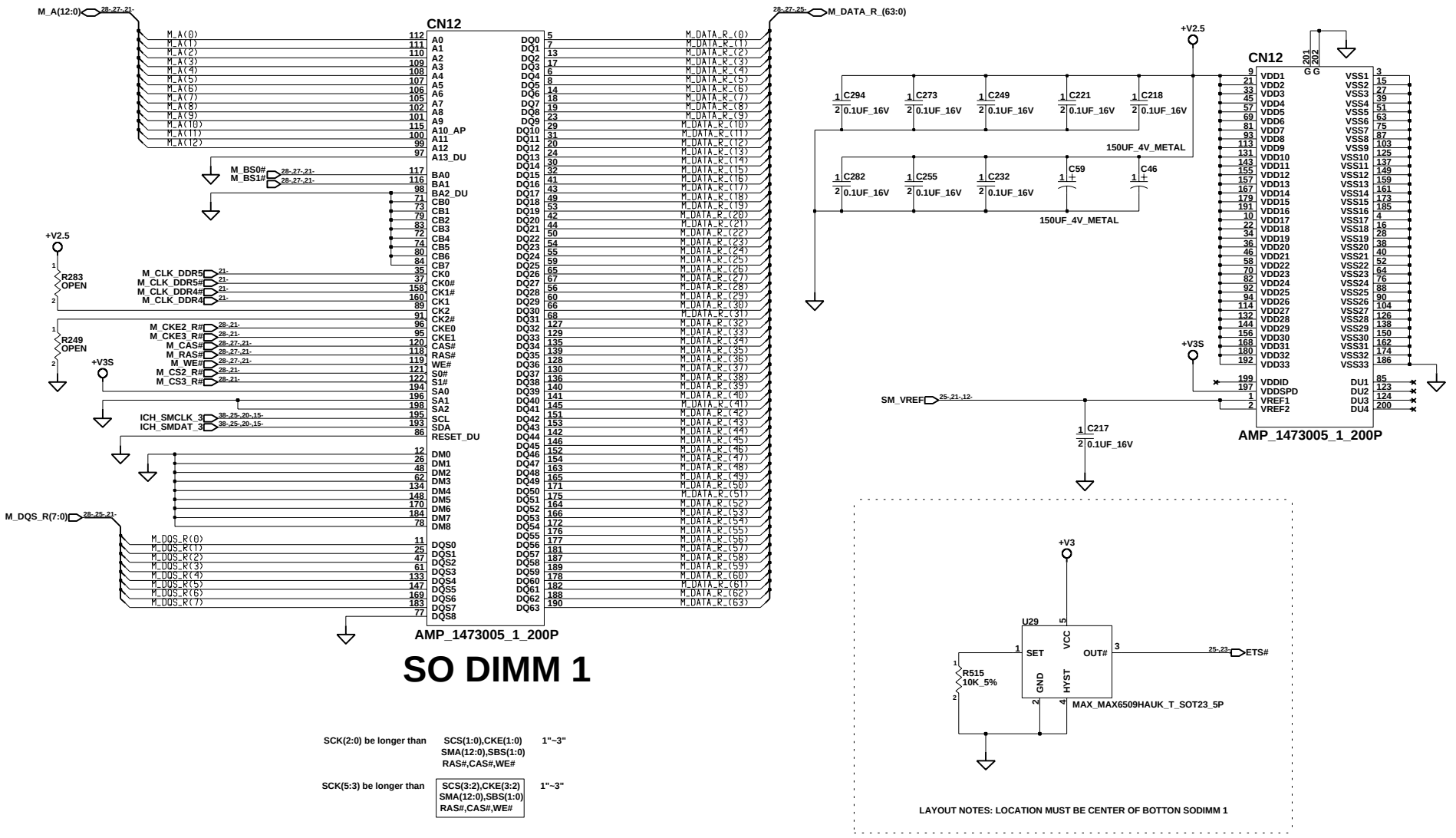
CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
DDR-SDRAM-1

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 25 OF 69



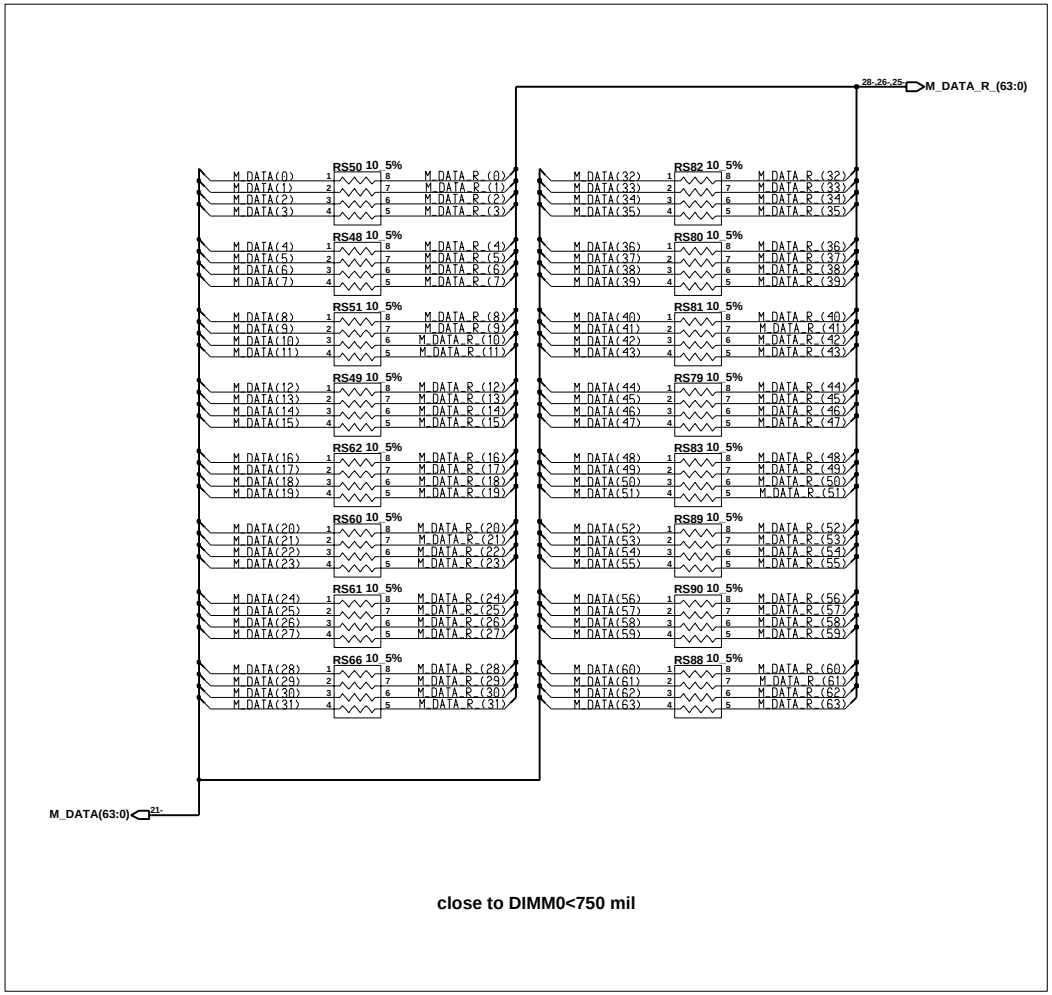
CHANGE by		
MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
DDR-SDRAM-2

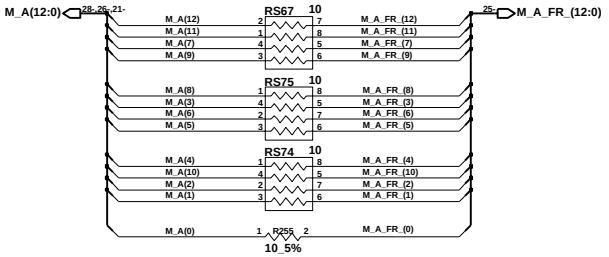
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 26 OF 69

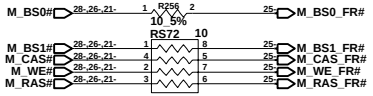


close to DIMM0<750 mil

close to DIMM1<750 mil



ALL RESISTORS ARE 10 ohm



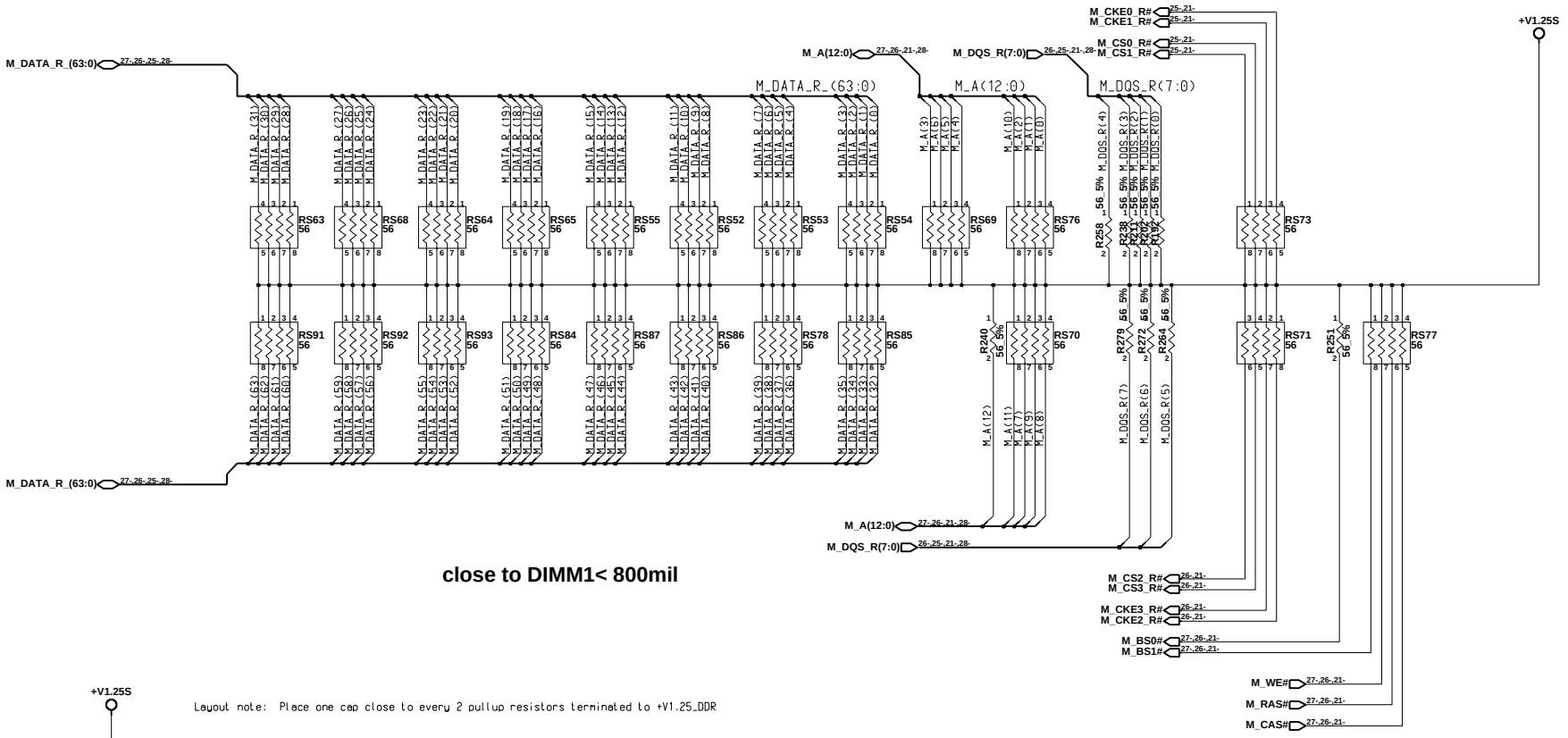
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

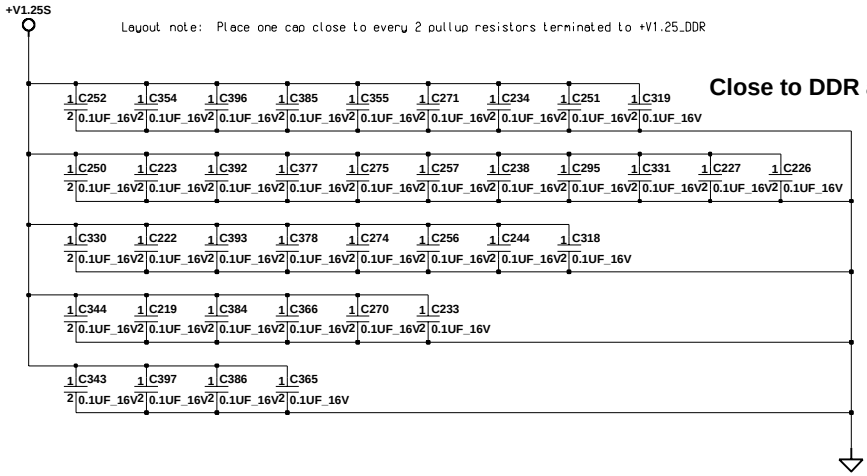
INVENTEC

TITLE: ***MODULES, RUBY1.0***
DDR-SDRAM-3

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 27 OF 69





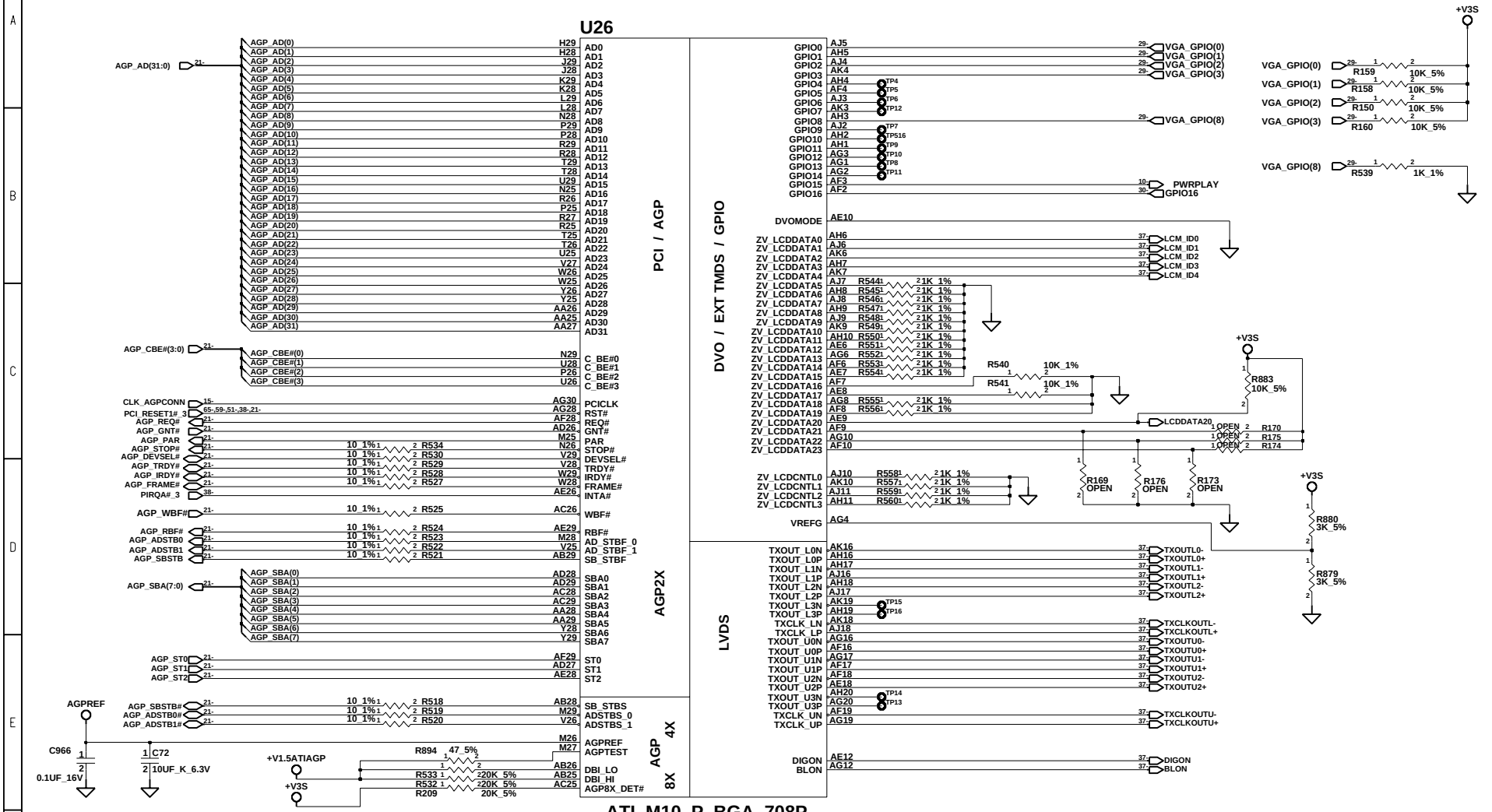
CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
DDR-SDRAM-4

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

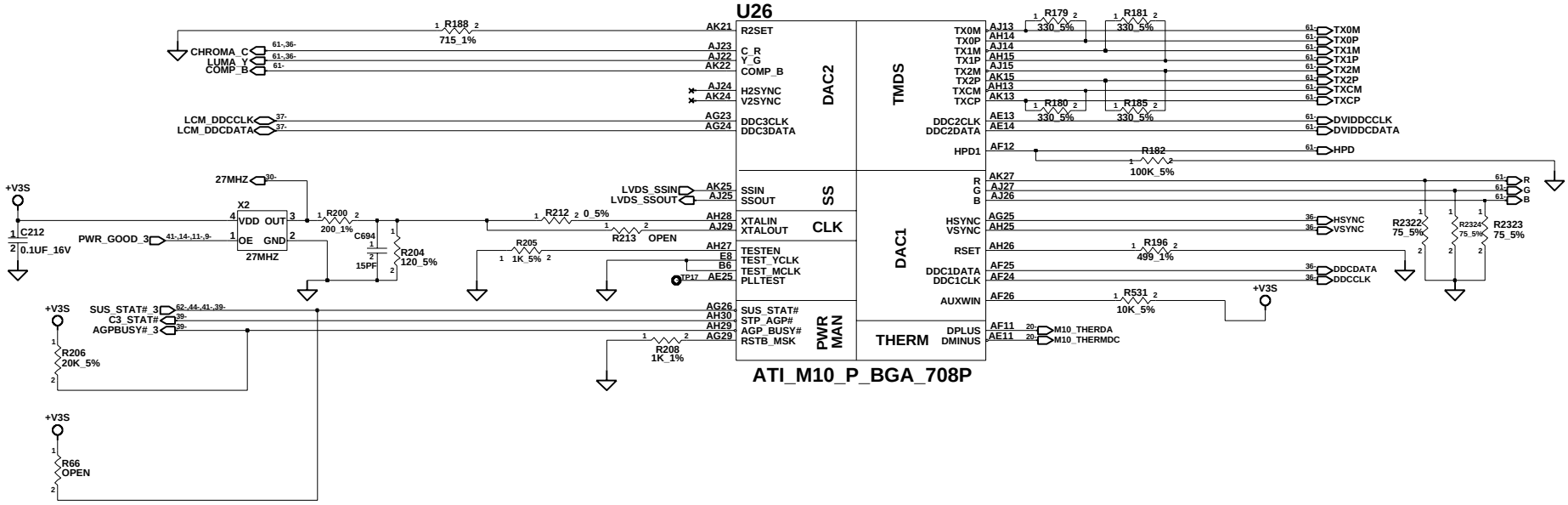
SHEET 28 OF 69

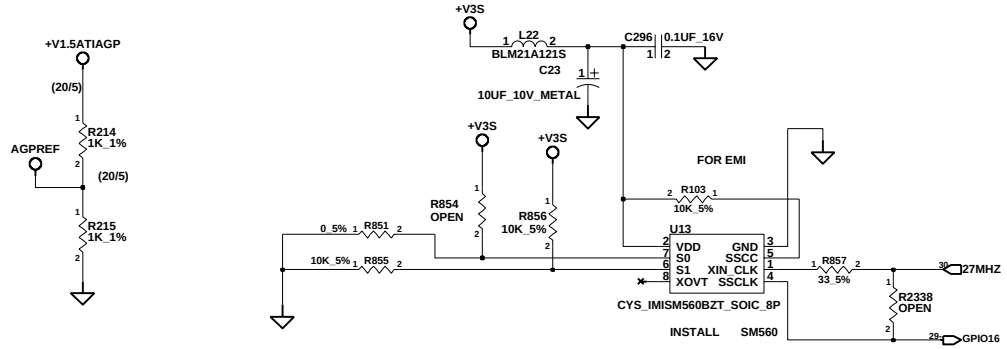


TITLE	MODULES, RUBY1.0
	ATI-M10-P-1
SIZE	A3
CODE	CS
DOC. NUMBER	131MBD2600-0-0
REV	D
SHEET	29 OF 69
CHANGE by	MORRIS CHANG
	23-Jul-2003

INVENTEC

LAYOUT NOTES : R1359, R1361, R1360, R1362 CLOSE TO M10-P
TERMINATIONS TO TMDS SIGNALS





CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

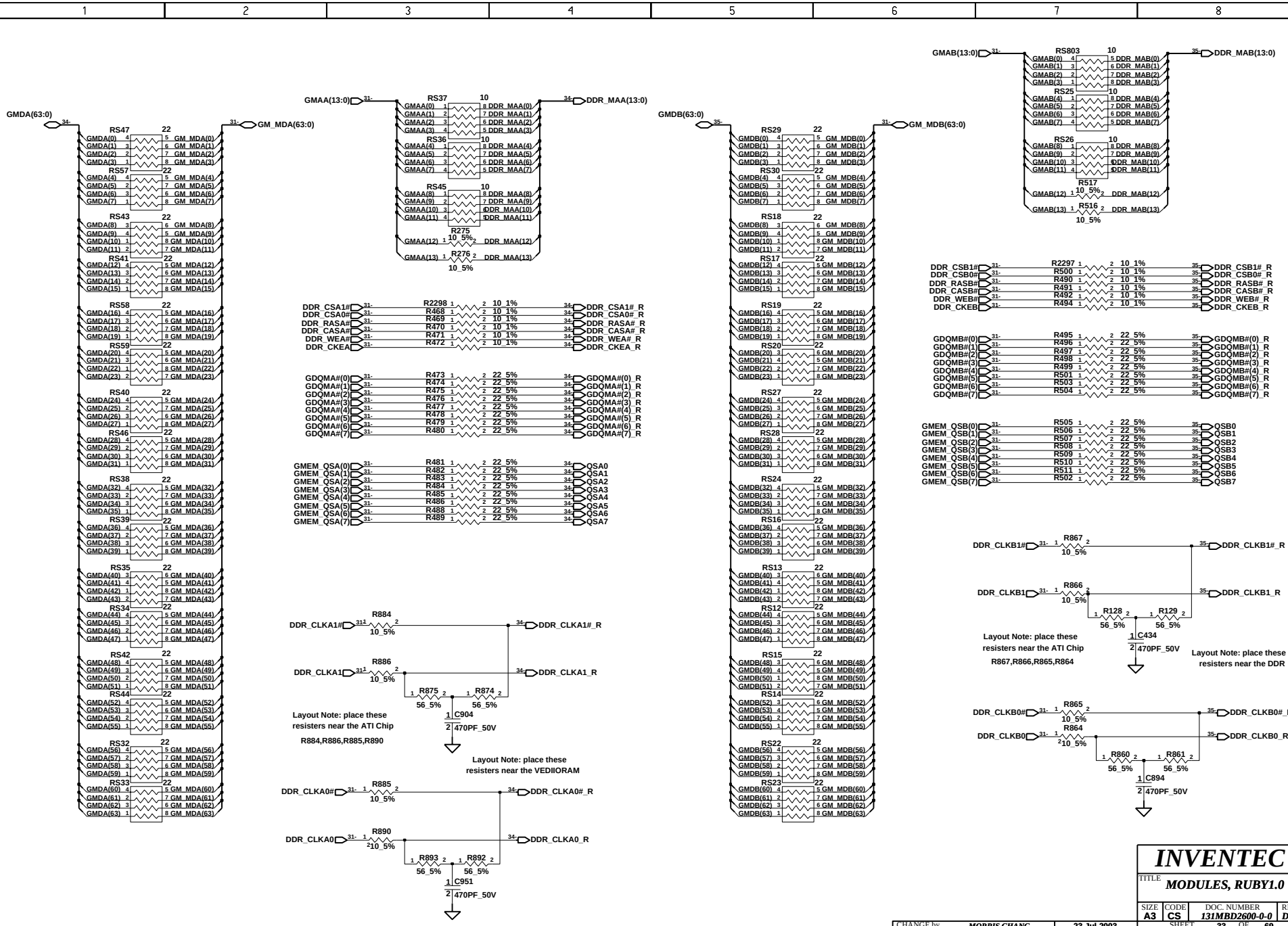
TITLE: *MODULES, RUBY1.0*
ATI-M10-P-2

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 30 OF 69

MEMORY INTERFACE A

MEMORY INTERFACE B



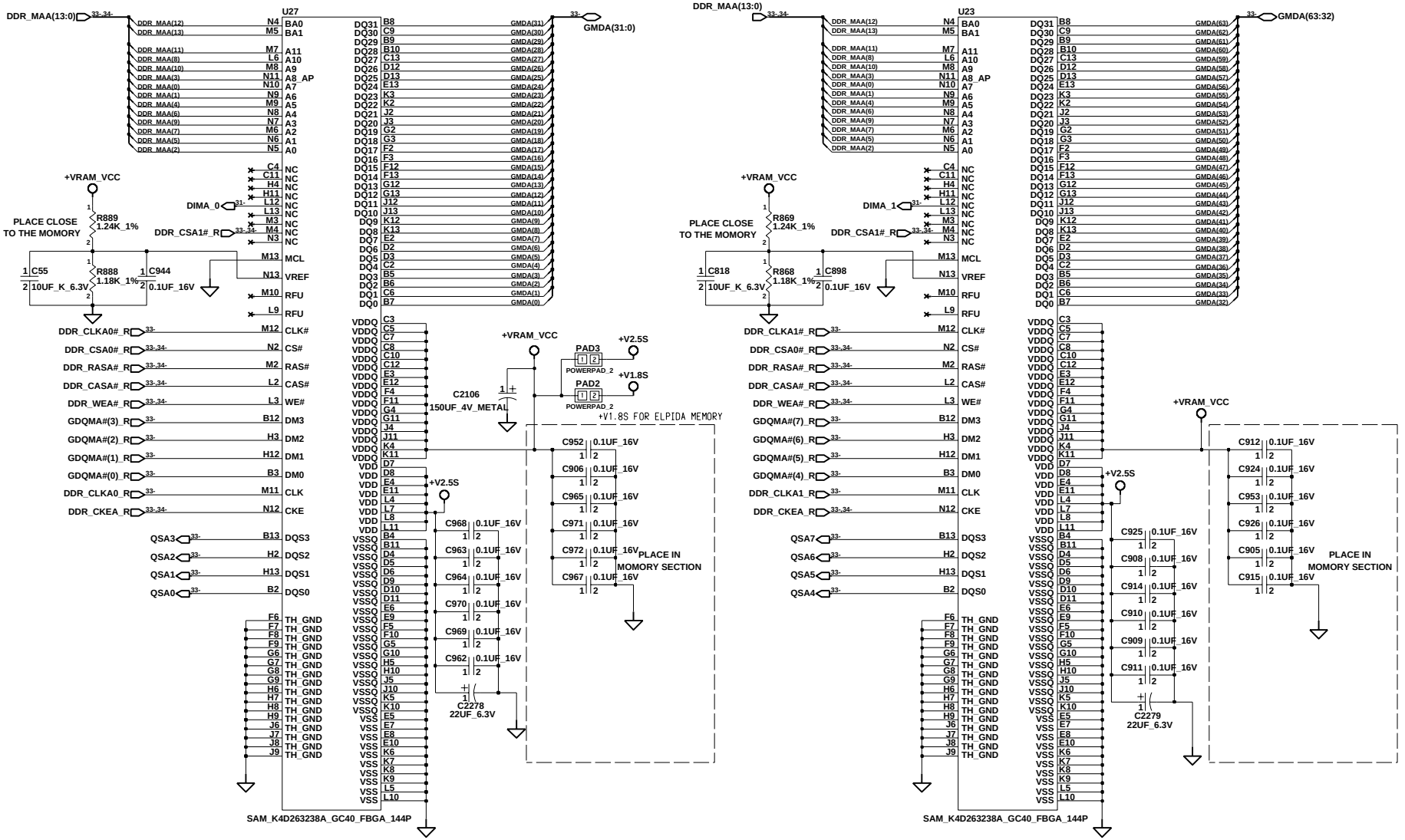
CHANGE by	*MORRIS CHANG*	*23-Jul-2003*	

INVENTEC

TITLE: ***MODULES, RUBY1.0***

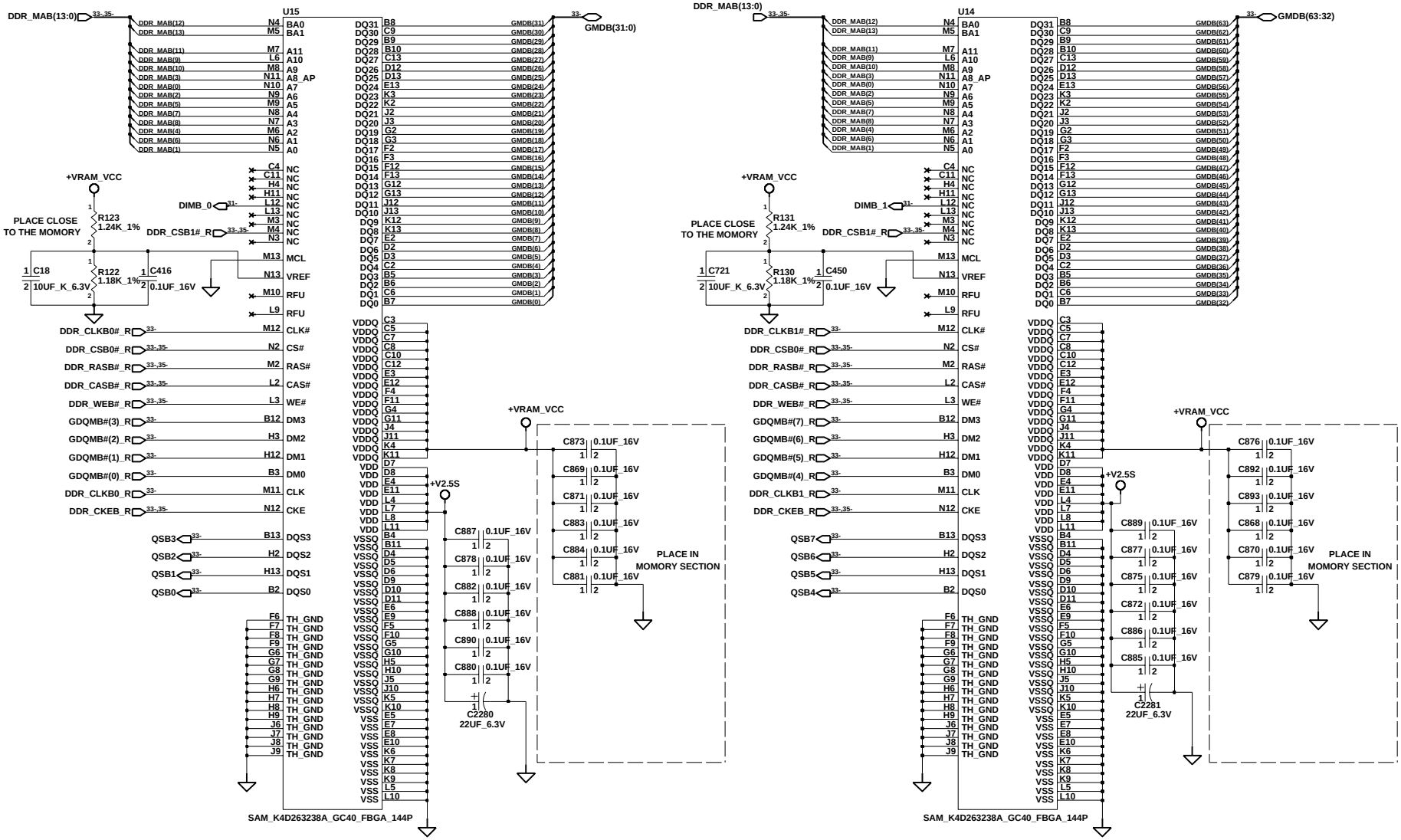
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 33 OF 69

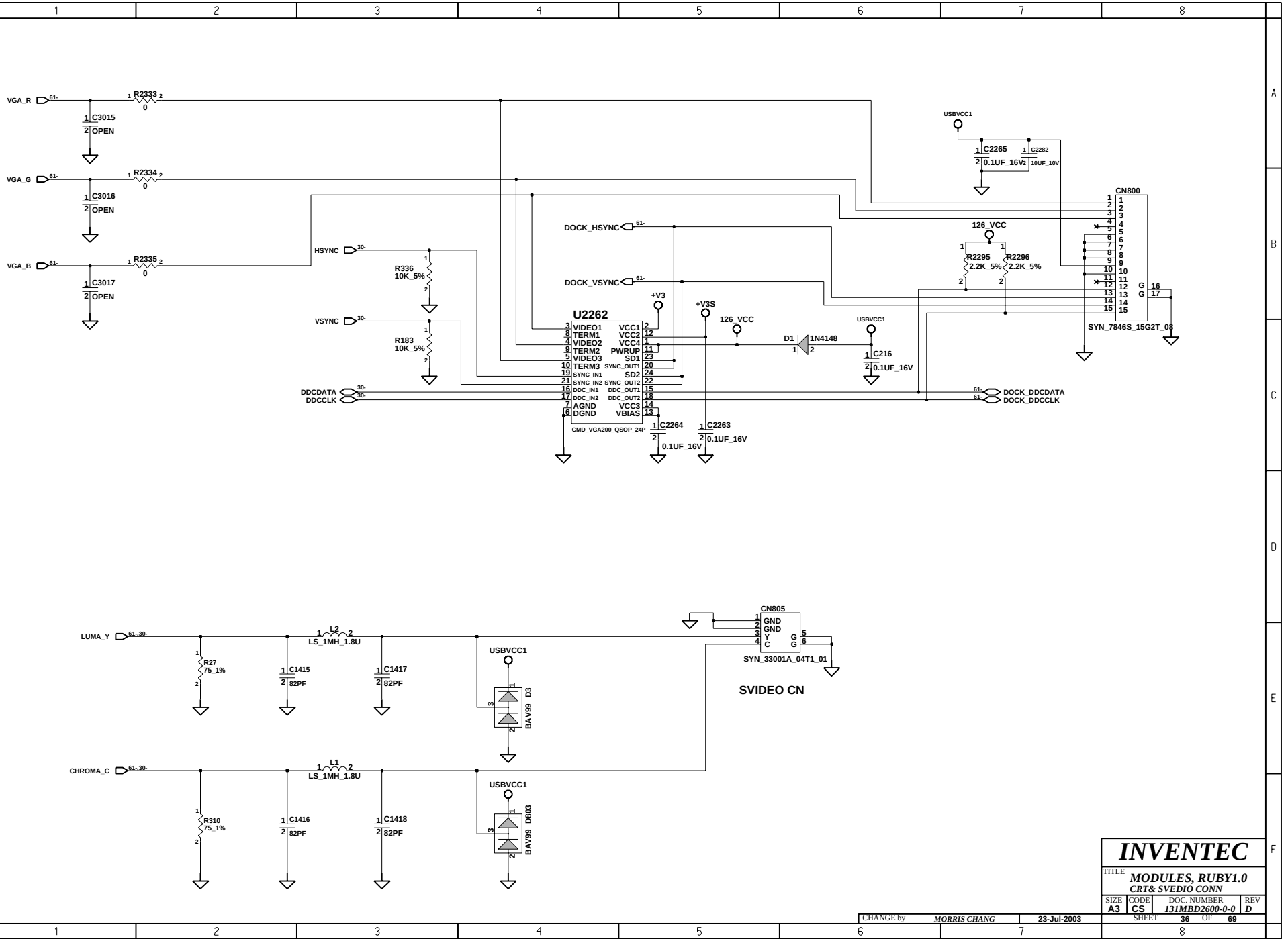


TITLE	*MODULES, RUBY1.0*
SIZE	A3
CODE	CS
DOC. NUMBER	*131MBD2600-0-0*
REV	*D*
SHEET	34 OF 69
CHANGE by	*MORRIS CHANG*
	23-Jul-2003

INVENTEC



INVENTEC			
TITLE	MODULES, RUBY1.0 VIDEO RAM-2		
SIZE A3	CODE CS	DOC. NUMBER 131MBD2600-0-0	REV D
CHANGE by MORRIS CHANG	23-Jul-2003	SHEET 35 OF 69	



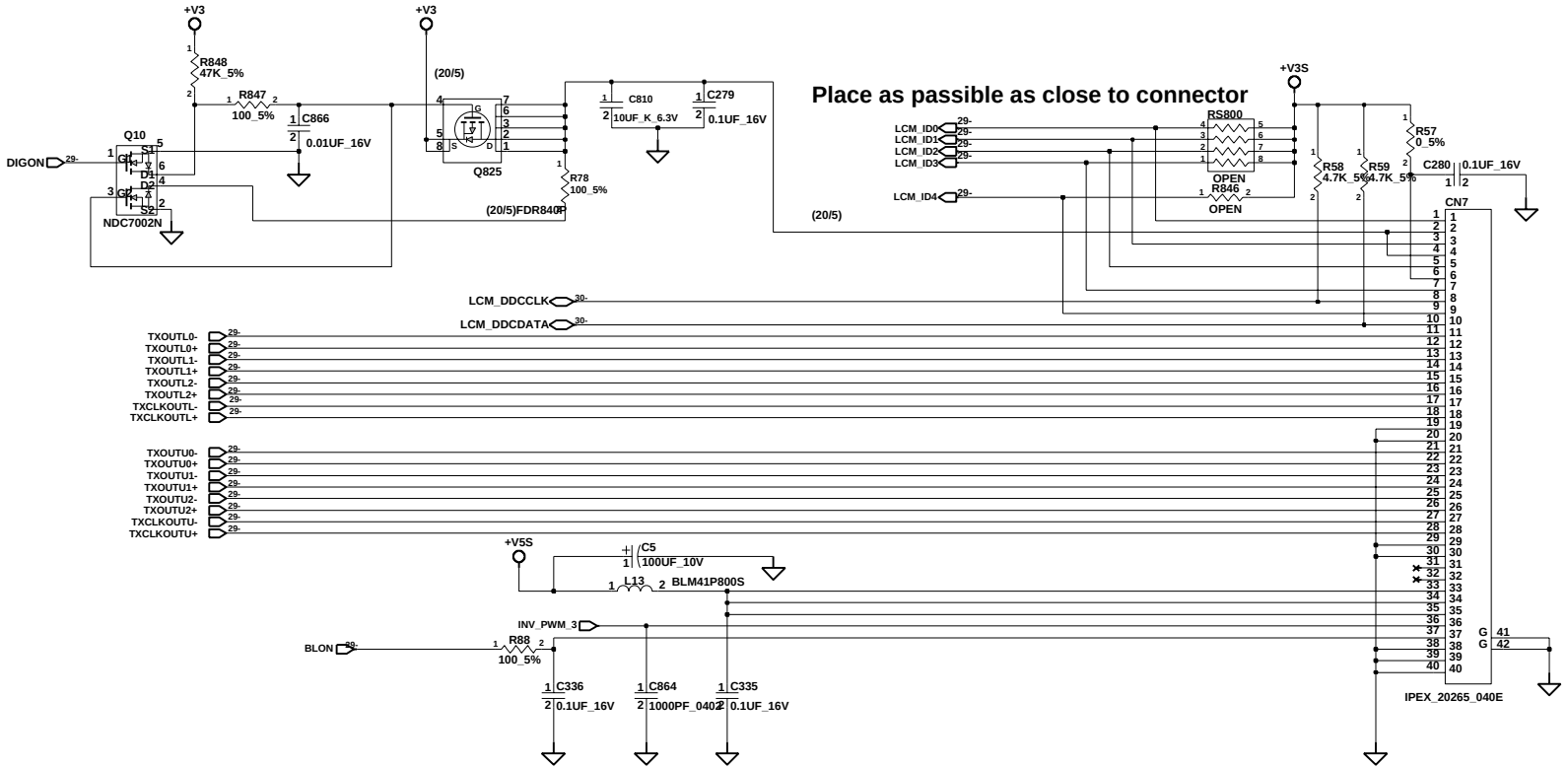
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
CRT& SVEDIO CONN

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 36 OF 69



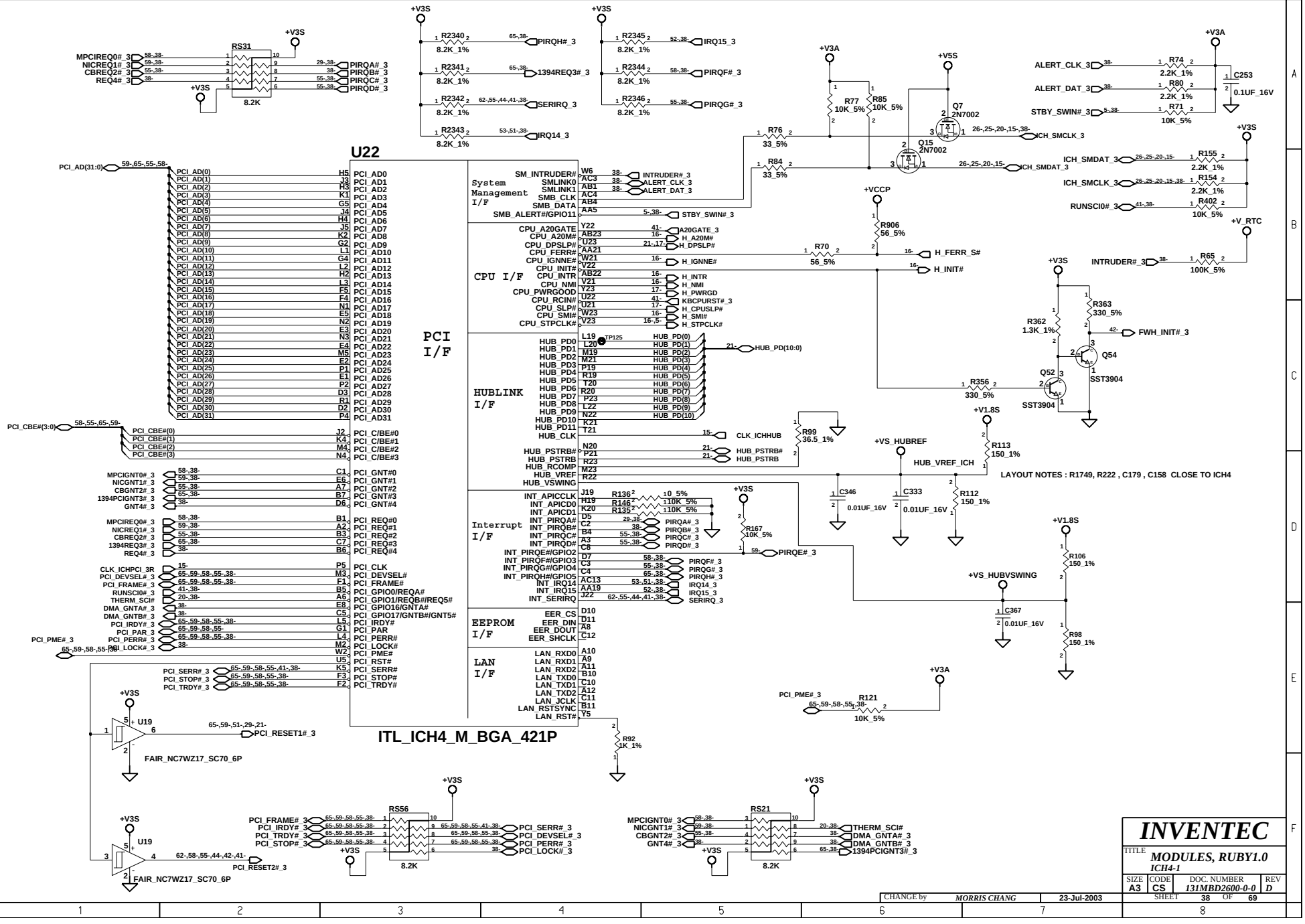
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
LCD CONN

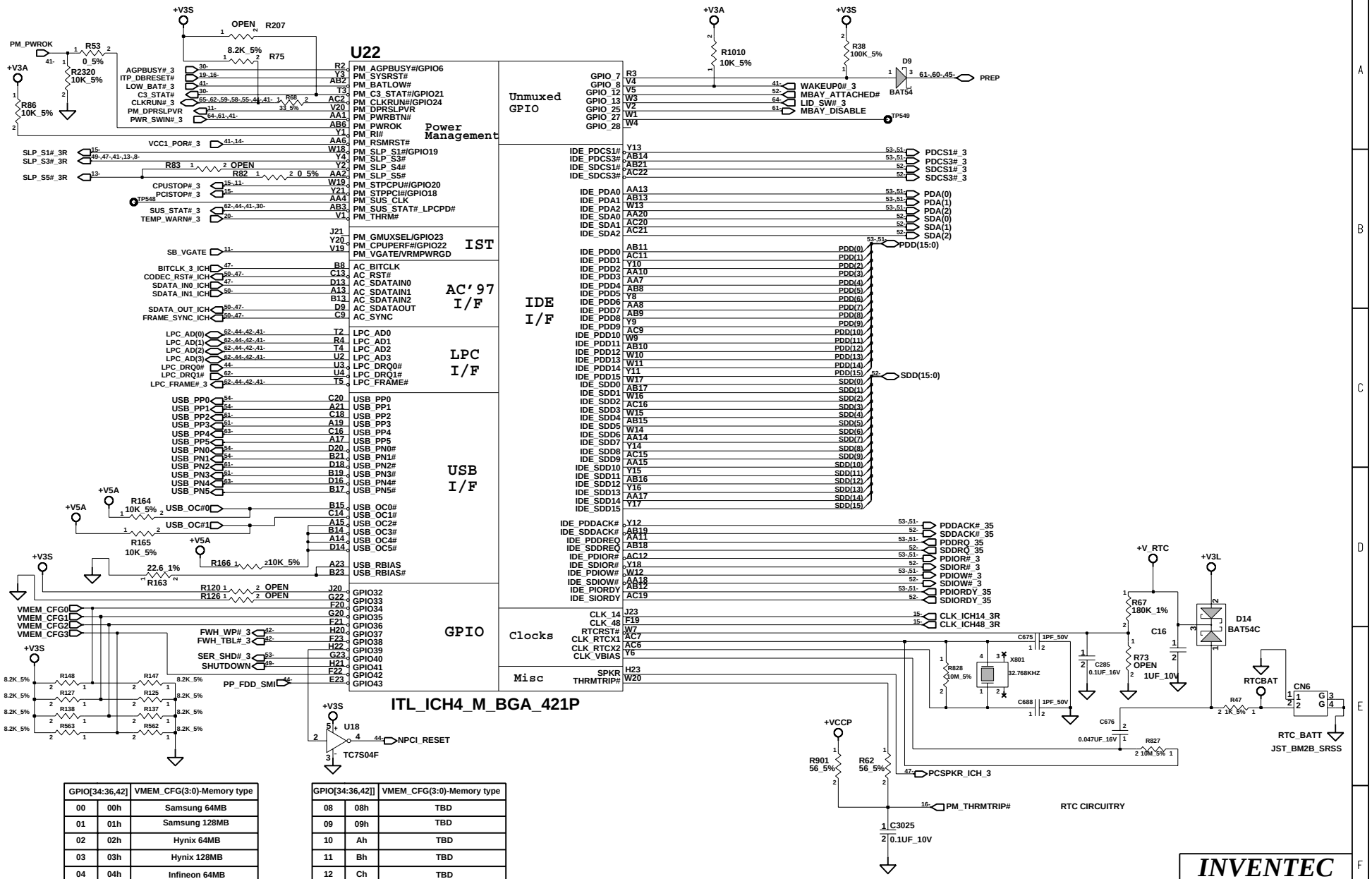
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 37 OF 69



TITLE	MODULES, RUBY1.0 ICH4-1
SIZE	A3
CODE	CS
DOC. NUMBER	131MBD2600-0-0
REV	D
SHEET	38 OF 69
CHANGE by	MORRIS CHANG
	23-Jul-2003

INVENTEC



GPIO[34:36,42]		VMEM_CFG(3:0)-Memory type
00	00h	Samsung 64MB
01	01h	Samsung 128MB
02	02h	Hynix 64MB
03	03h	Hynix 128MB
04	04h	Infineon 64MB
05	05h	Infineon 128MB
06	06h	TBD
07	07h	TBD

GPIO[34:36,42]]		VMEM_CFG(3:0)-Memory type
08	08h	TBD
09	09h	TBD
10	Ah	TBD
11	Bh	TBD
12	Ch	TBD
13	Dh	TBD
14	Eh	TBD
15	Fh	TBD

CHANGE by	MORRIS CHANG	23-Jul-2003	

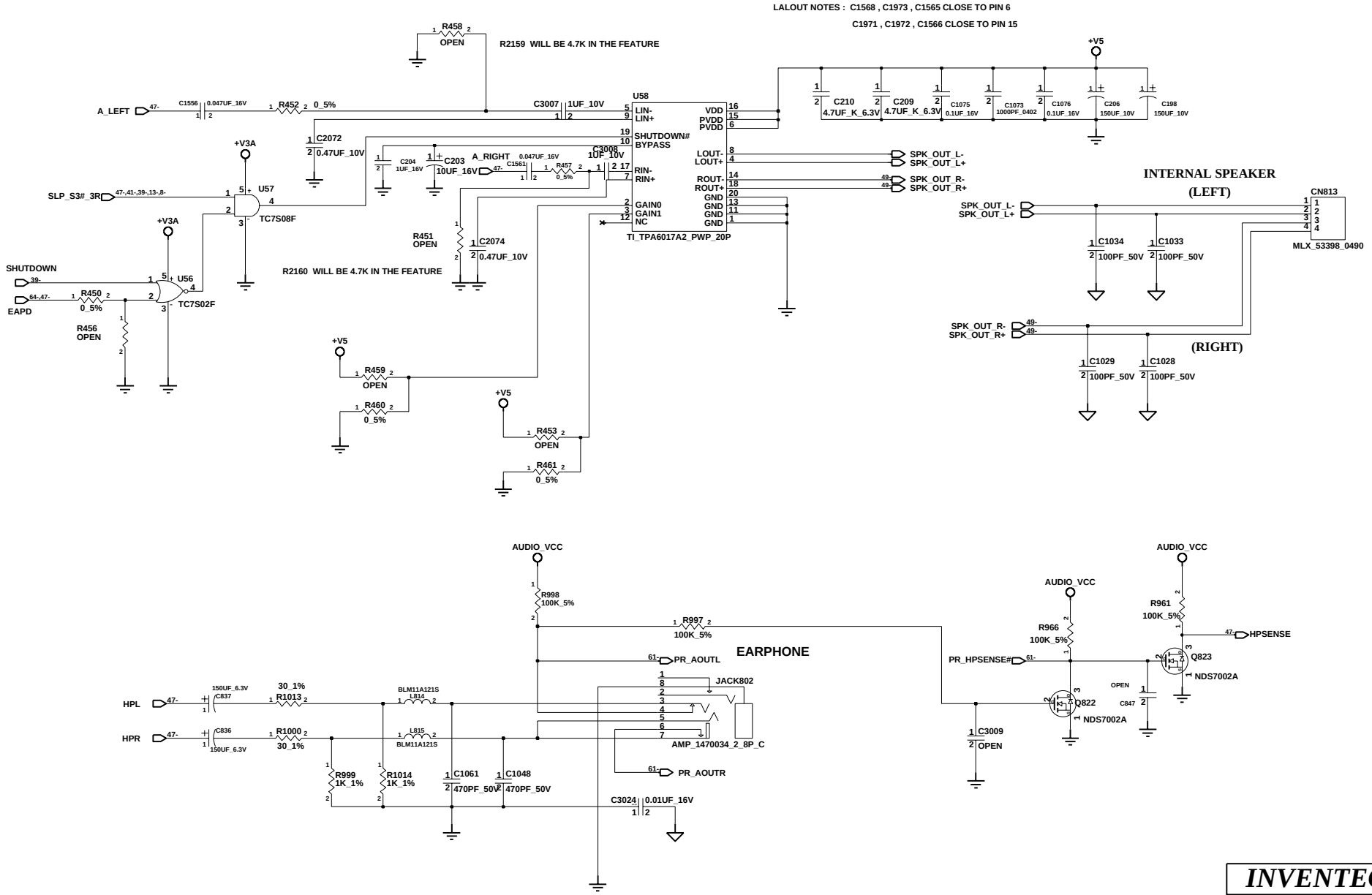
INVENTEC

TITLE: *MODULES, RUBY1.0 ICH4-2*

SIZE	CODE	DOC. NUMBER	REV
A3	CS	131MBD2600-0-0	D

SHEET 39 OF 69

VSS
POWER



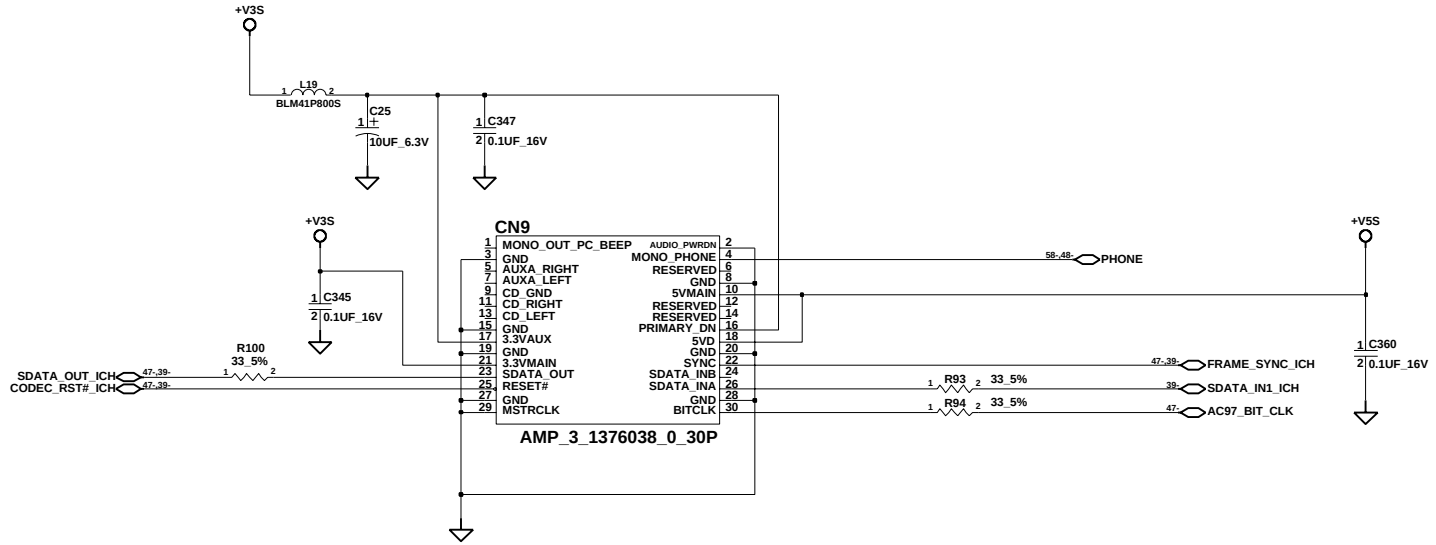
CHANGE by	*MORRIS CHANG*	23-Jul-2003		

INVENTEC

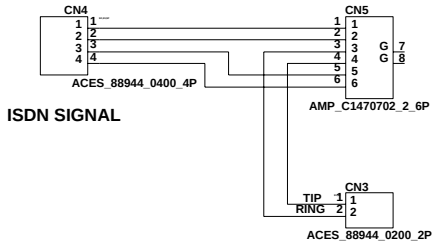
TITLE: ***MODULES, RUBY1.0***
AUDIO AMP & HP JACK

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 49 OF 69



MDC CONN



ISDN SIGNAL

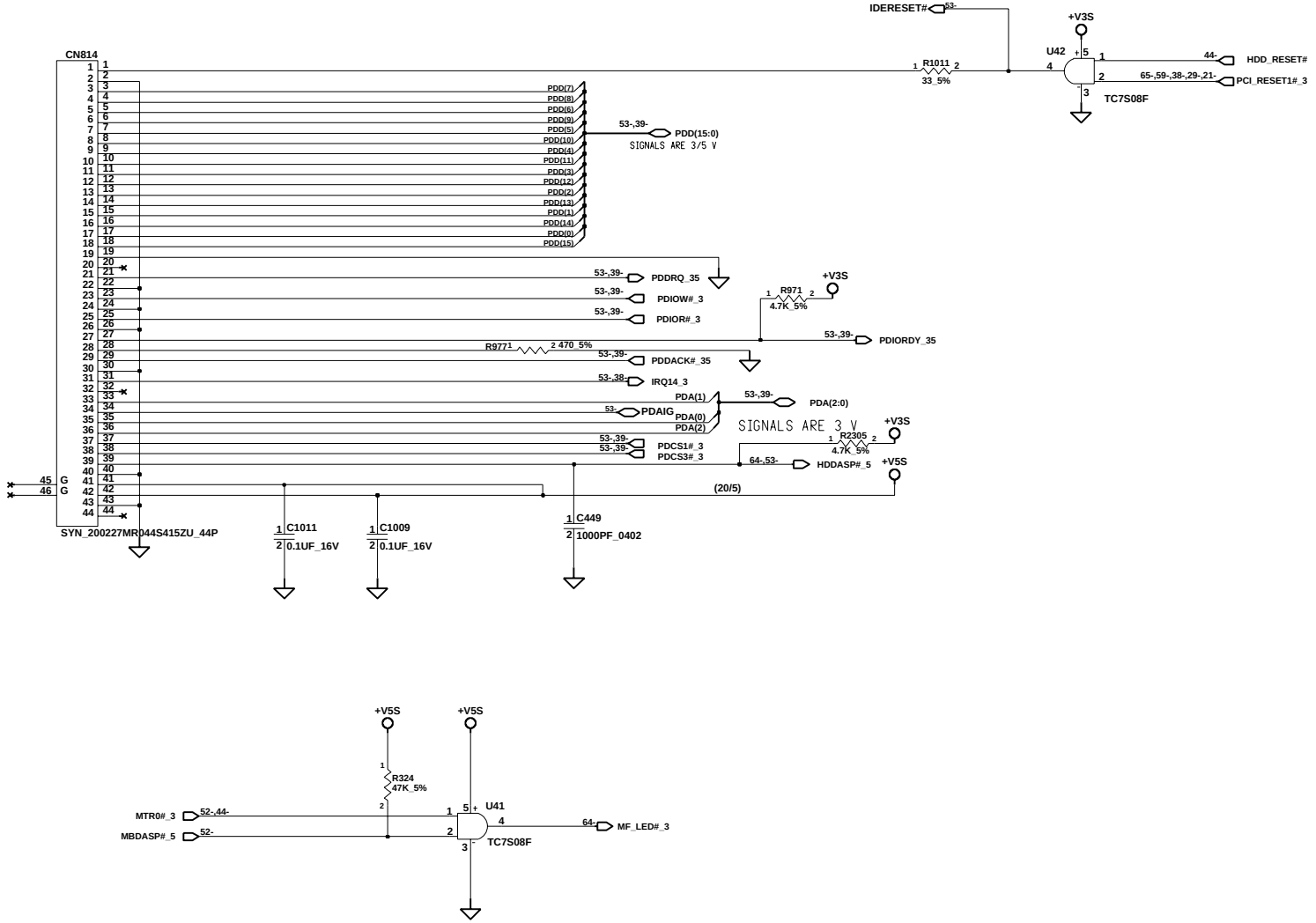
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
MDC CONN

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

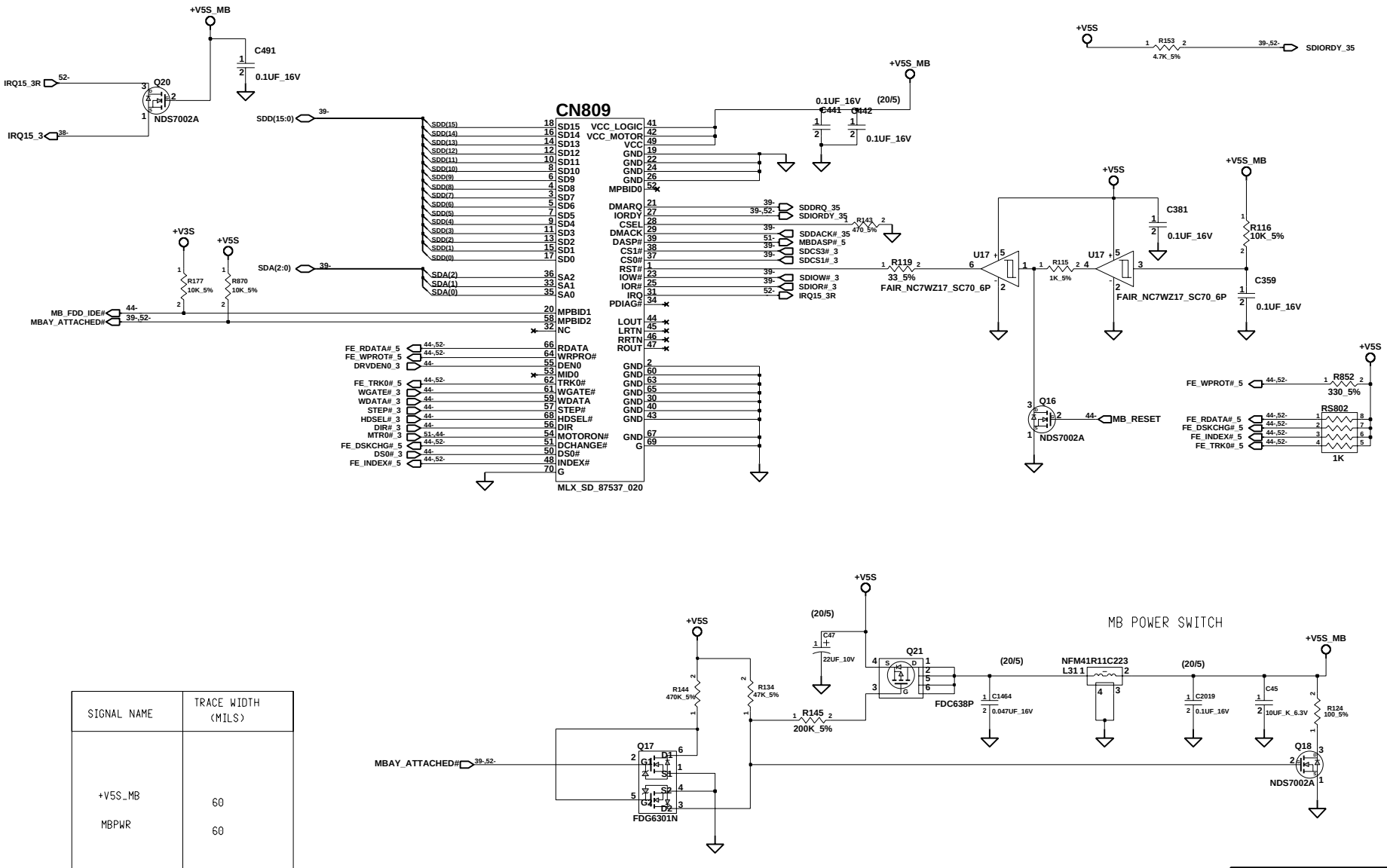
SHEET 50 OF 69



TITLE	*MODULES, RUBY1.0*
	HDD CONN
SIZE	A3
CODE	CS
DOC. NUMBER	*131MBD2600-0-0*
REV	*D*
SHEET	51 OF 69

INVENTEC

CHANGE by		
MORRIS CHANG	23-Jul-2003	



SIGNAL NAME	TRACE WIDTH (MILS)
+V5S_MB	60
MBPWR	60

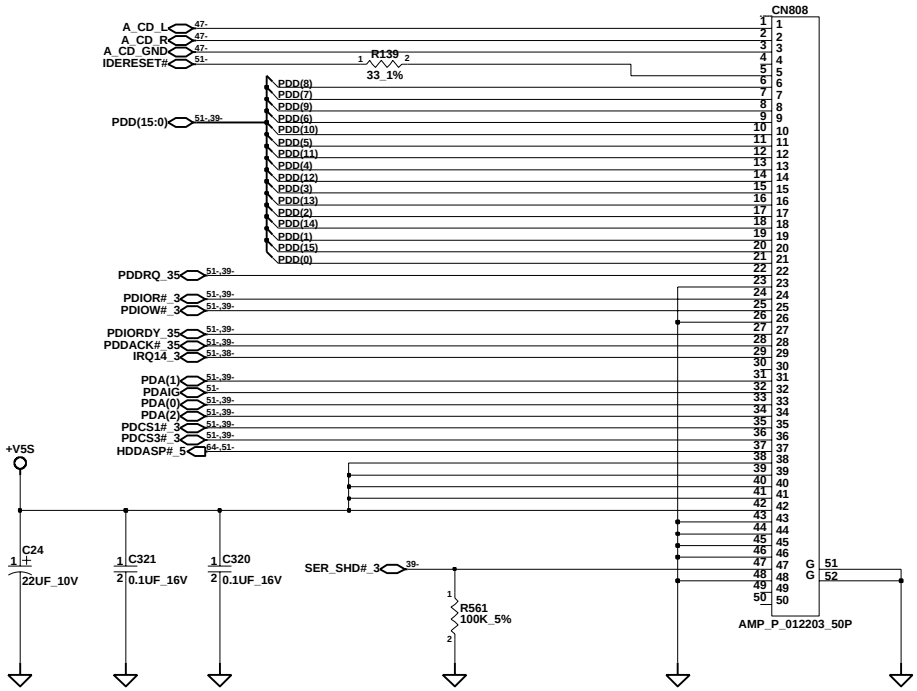
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
MULTIBAY CONN

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 52 OF 69



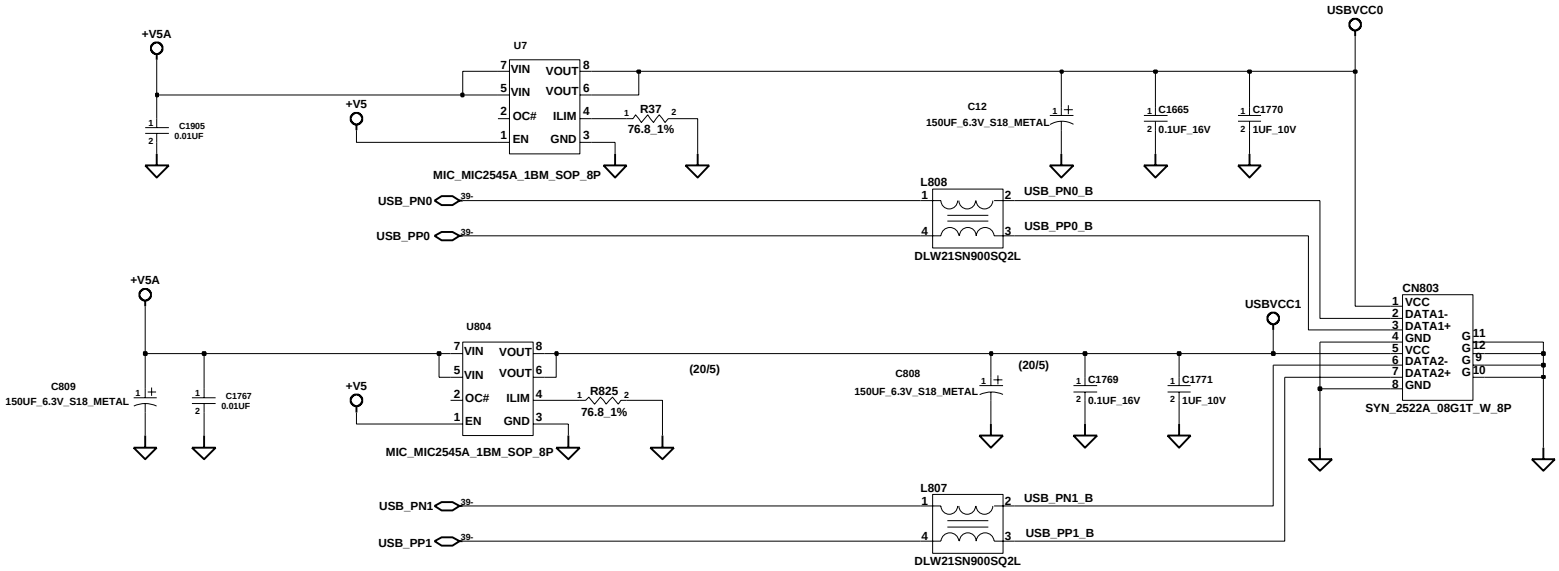
CSEL	Drive address
GND	0
OPEN	1

CSEL=47PIN

CHANGE by	*MORRIS CHANG*	23-Jul-2003	
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*
SHEET	53	OF	69

INVENTEC

TITLE
MODULES, RUBY1.0
SELECT BAY I/F



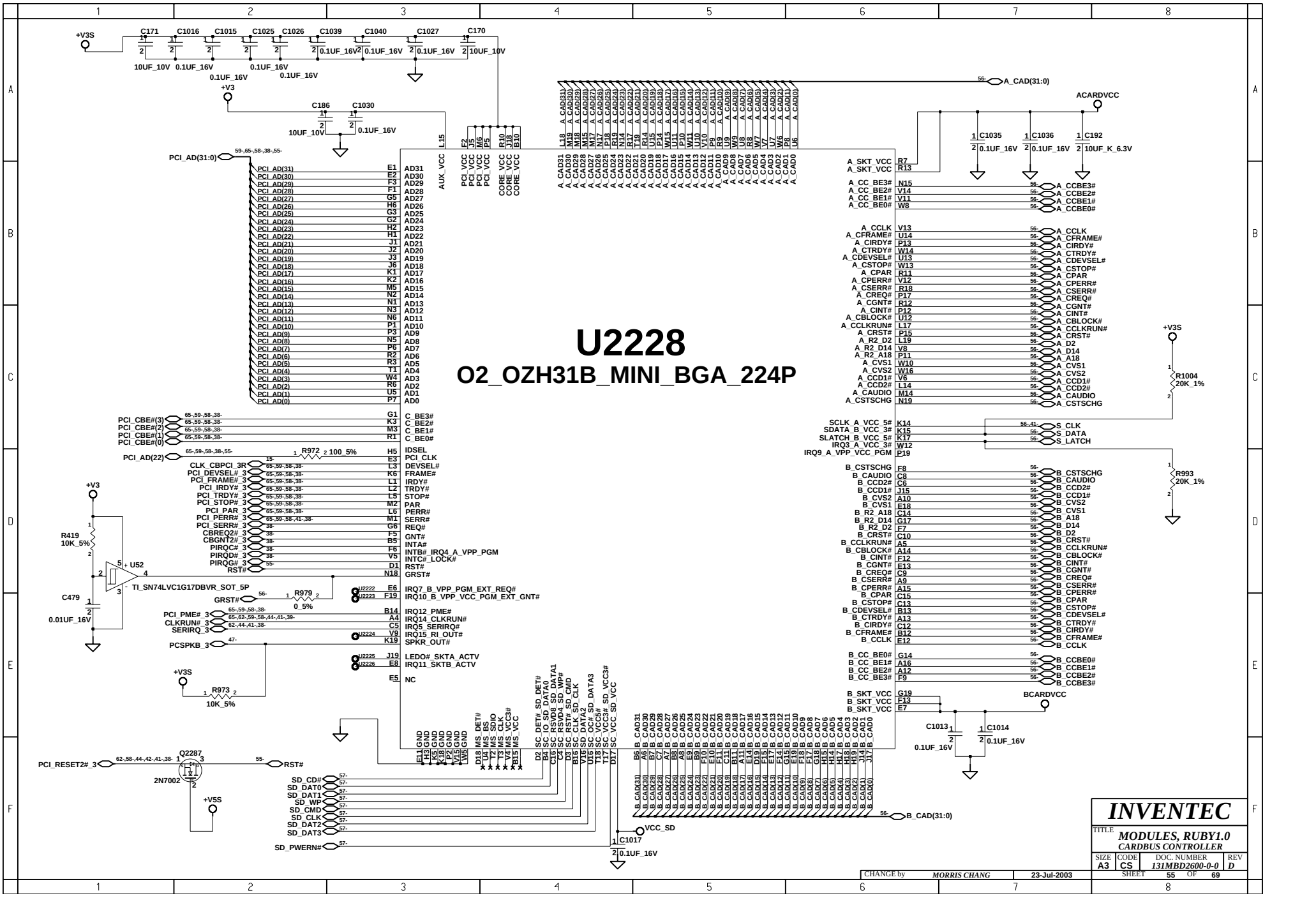
CHANGE by	*MORRIS CHANG*	23-Jul-2003		

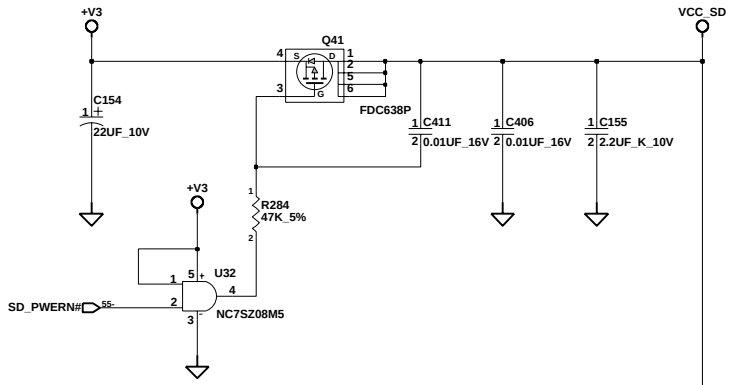
INVENTEC

TITLE: ***MODULES, RUBY1.0***
USB&IR CONN

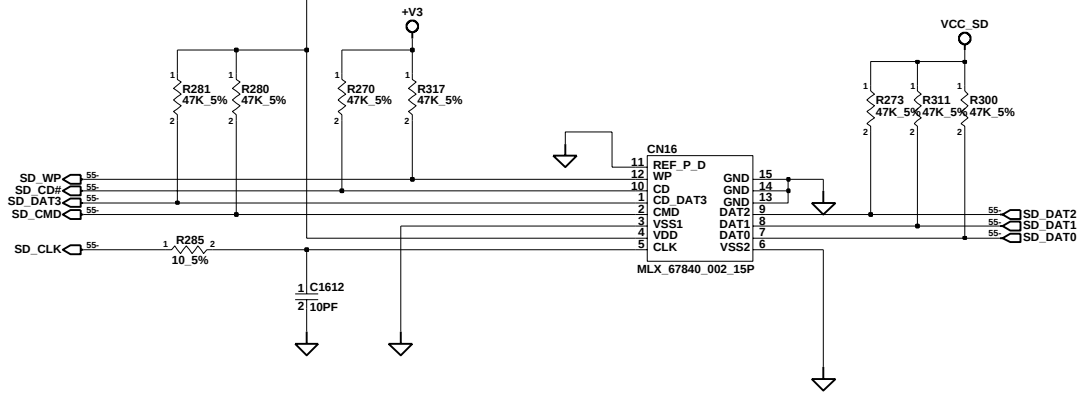
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 54 OF 69


U2228
O2_OZH31B_MINI_BGA_224P
INVENTEC
MODULES, RUBY1.0
CARDBUS CONTROLLER
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 55 OF 69
CHANGE by MORRIS CHANG
23-Jul-2003



SD CARD POWER



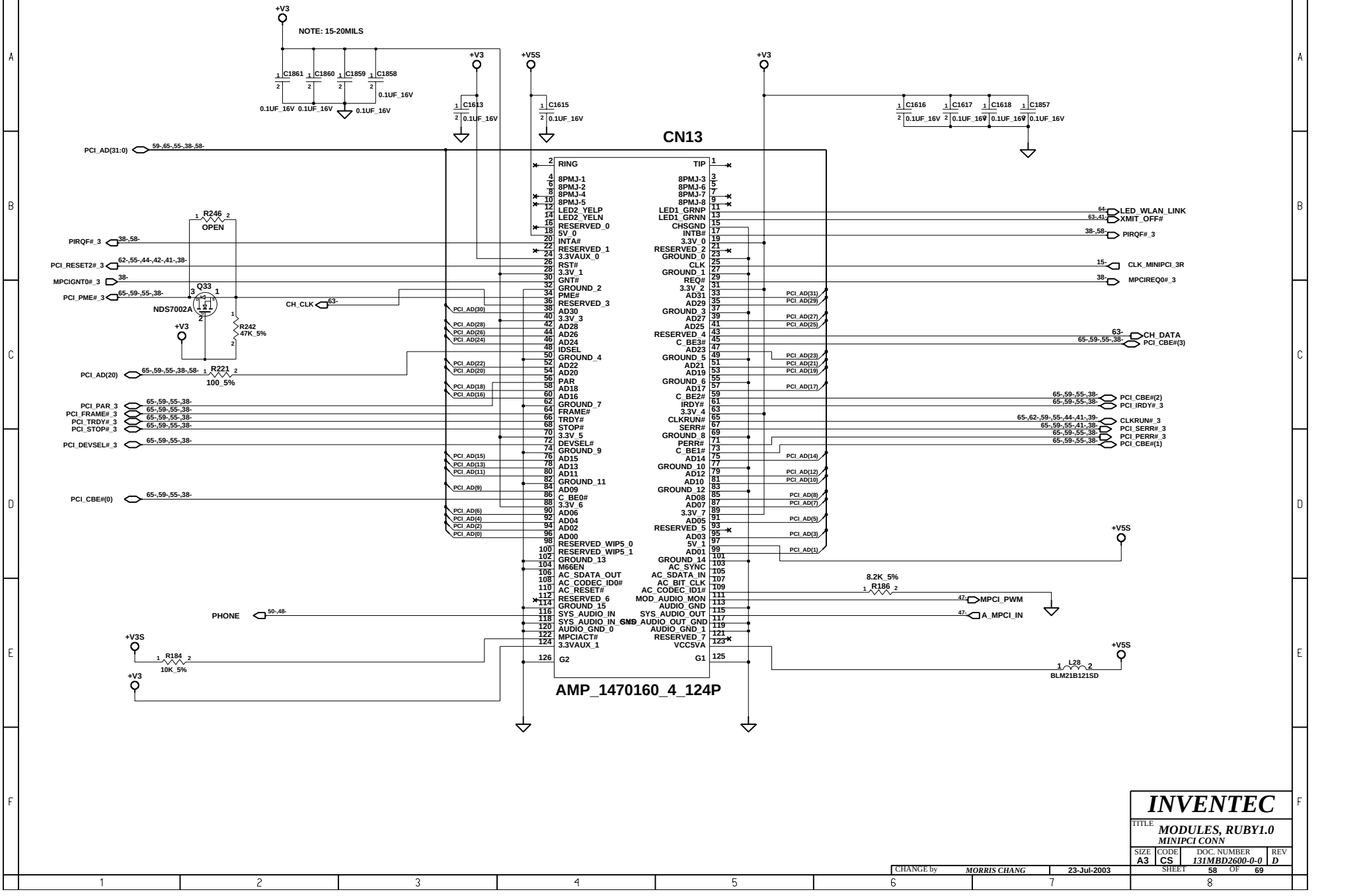
CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE: ***MODULES, RUBY1.0***
SD CARD CONN

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 57 OF 69



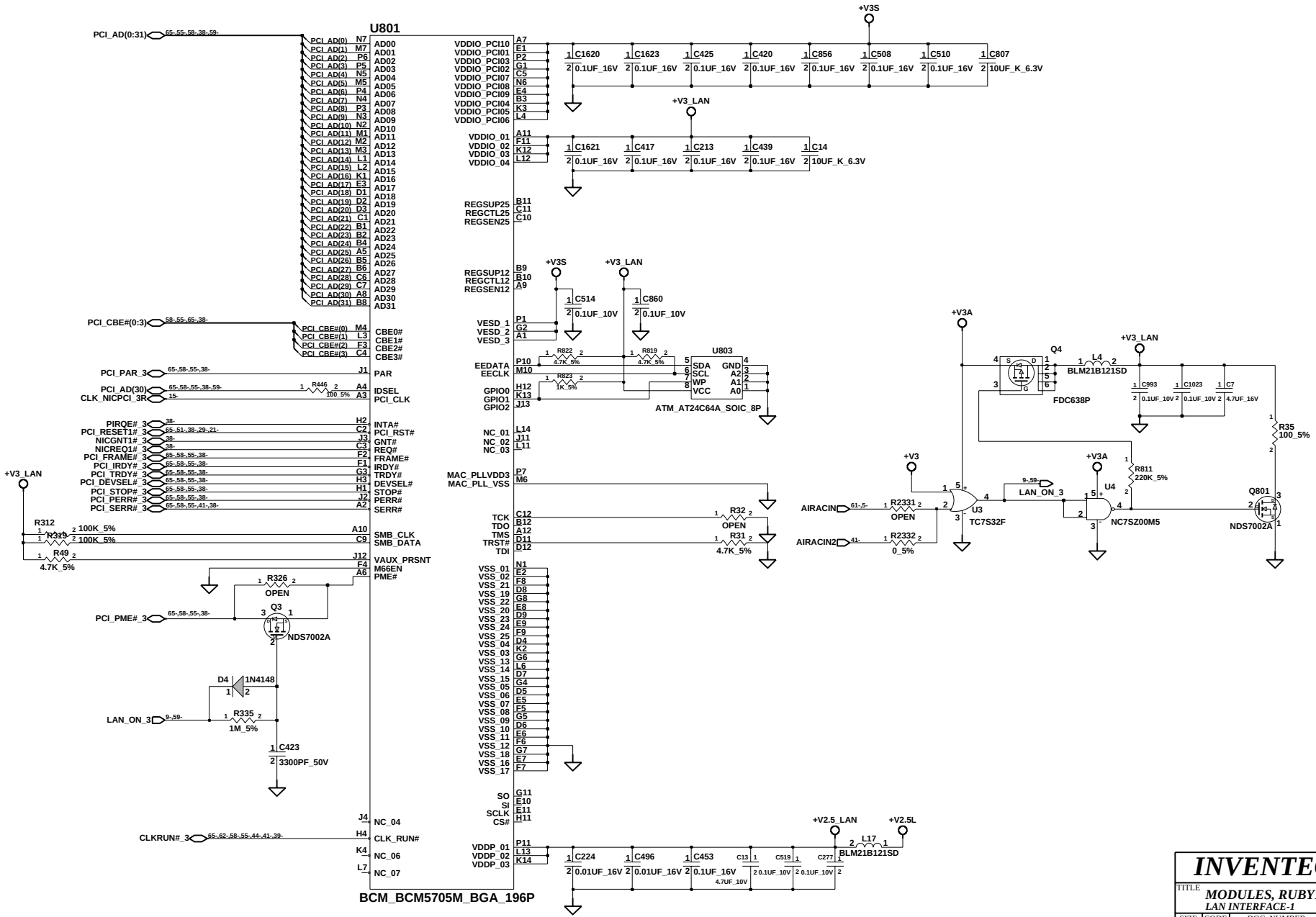
CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

TITLE: *MODULES, RUBY1.0*
MINIPCI CONN

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 58 OF 69



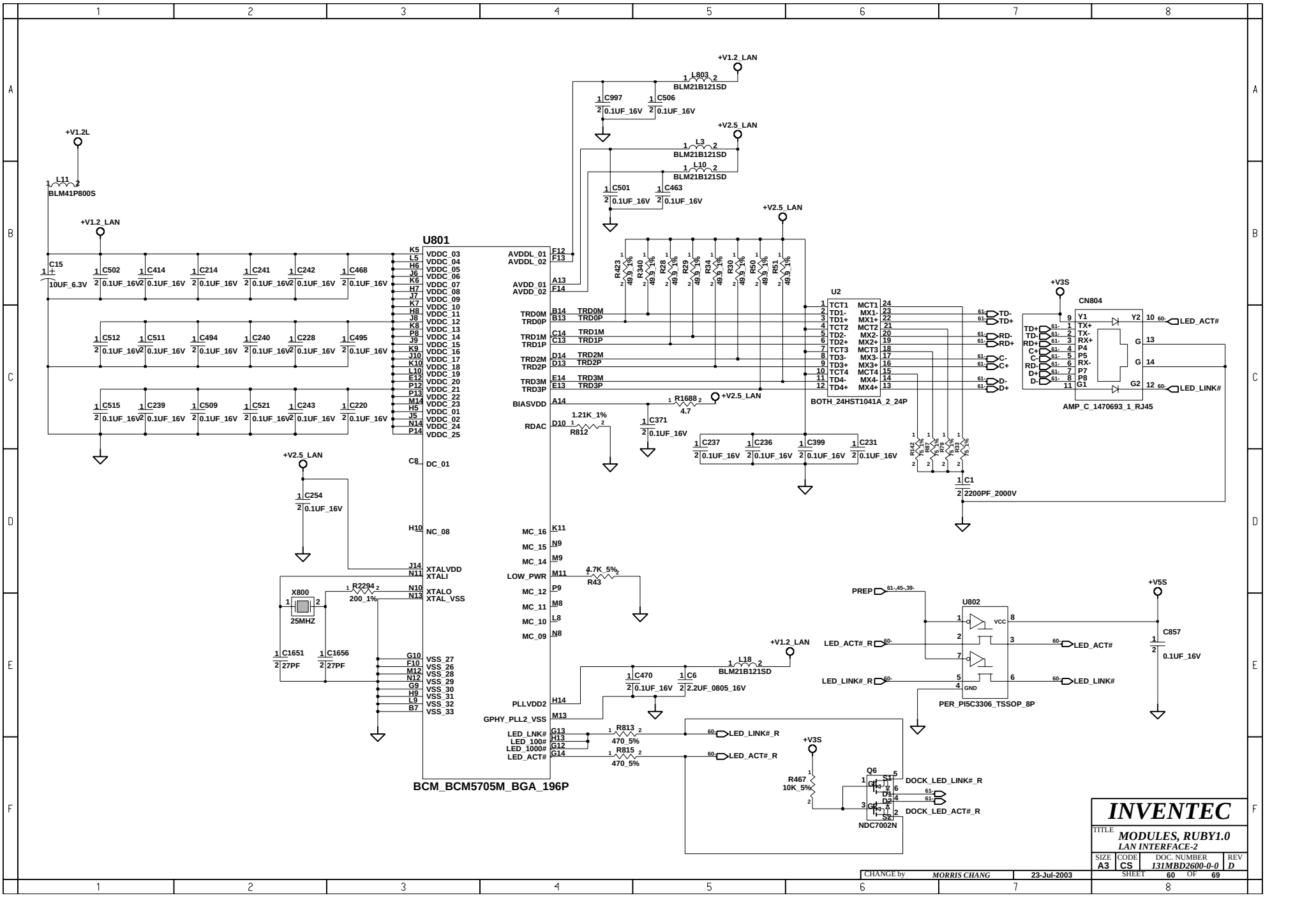
CHANGE by	MORRIS CHANG	23-Jul-2003	

INVENTEC

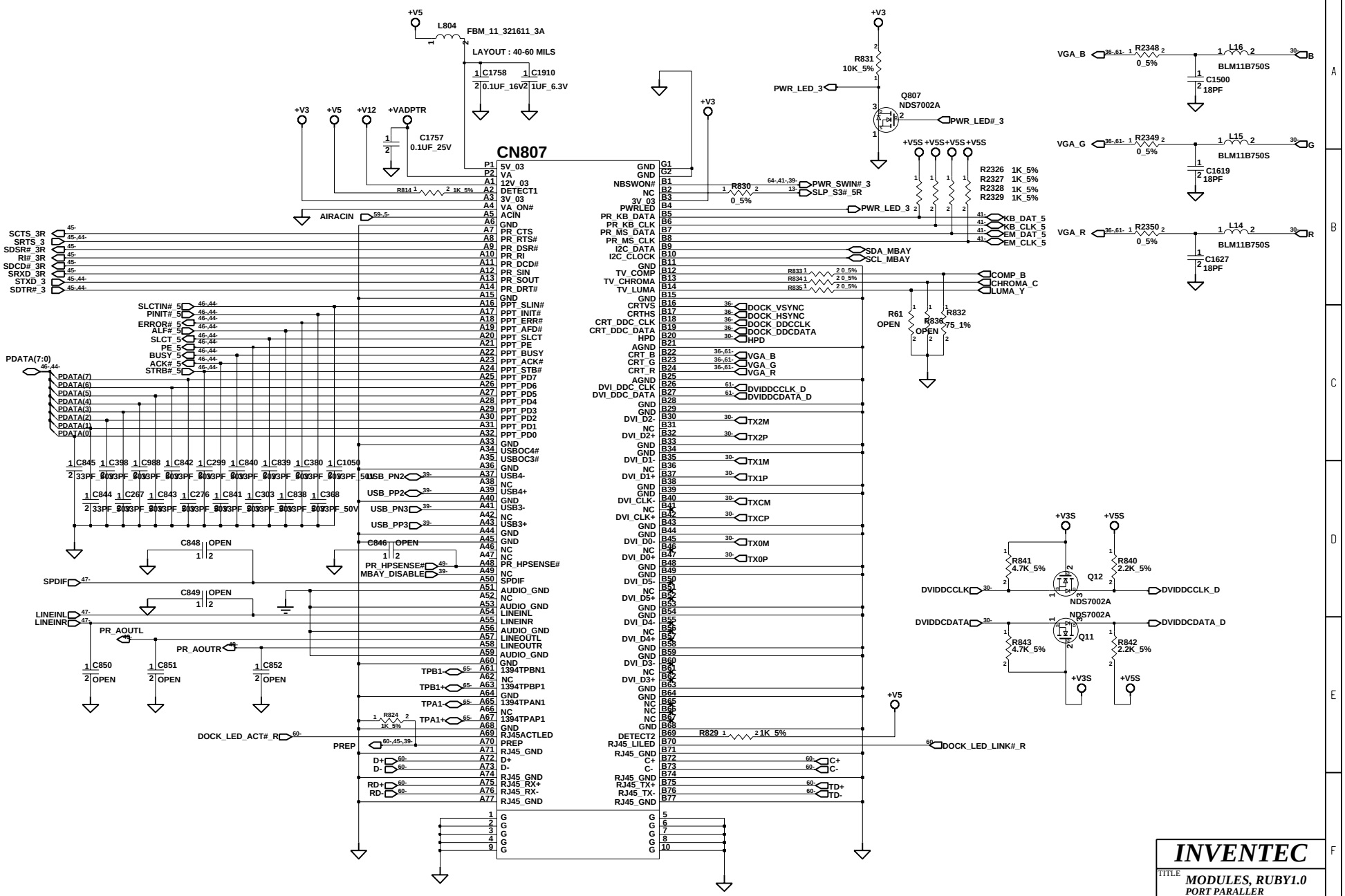
TITLE: *MODULES, RUBY1.0*
LAN INTERFACE-1

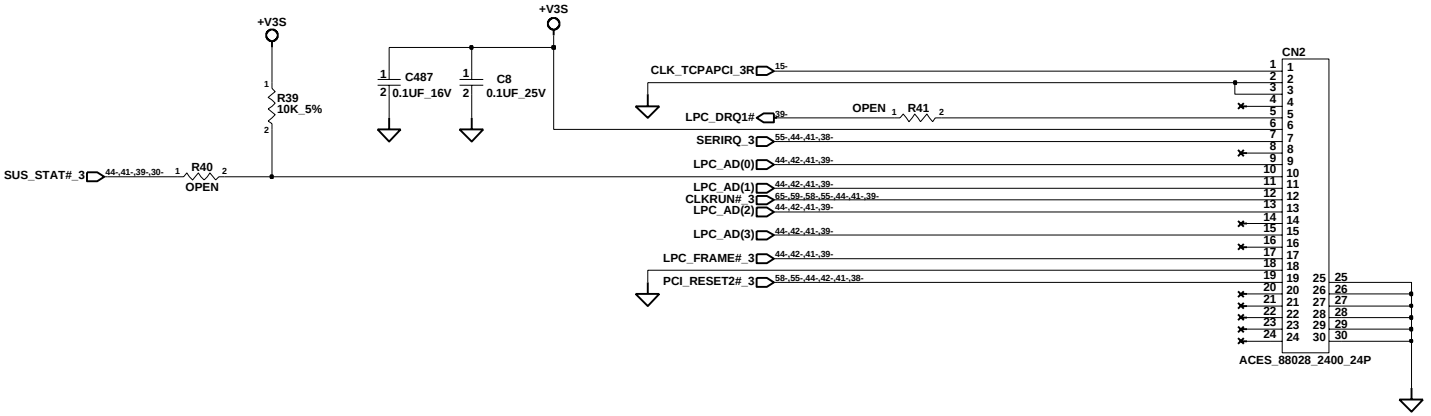
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 59 OF 69



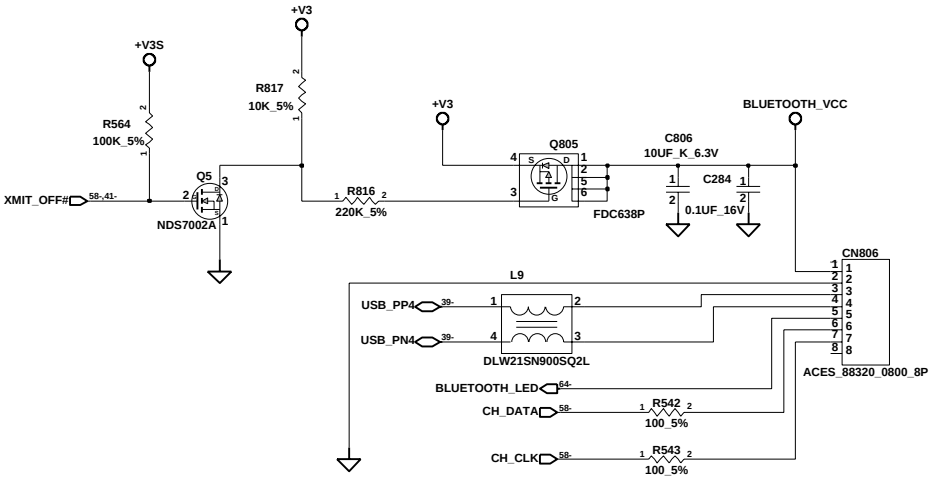
INVENTEC	
TITLE	*MODULES, RUBY1.0*
	LAN INTERFACE-2
SIZE	A3
CODE	CS
DOC. NUMBER	*131MBD2600-0-0*
REV	*D*
SHEET	60 OF 69
CHANGE by	*MORRIS CHANG*
	23-Jul-2003


CN807
JAE_WD_154S4VH_VF85_N_154P
L804
FBM_11_321611_3A
LAYOUT : 40-60 MILS
C1757
0.1UF_25V
R831
10K_5%
Q807
NDS7002A
PWR_LED_3
PWR_LED#_3
R2326 1K_5%
R2327 1K_5%
R2328 1K_5%
R2329 1K_5%
R2348
0_5%
L16
BLM11B750S
C1500
18PF
R2349
0_5%
L15
BLM11B750S
C1619
18PF
R2350
0_5%
L14
BLM11B750S
C1627
18PF
R61 OPEN
R832 75_1%
R836 OPEN
R841 4.7K_5%
R840 2.2K_5%
Q12
NDS7002A
R843 4.7K_5%
R842 2.2K_5%
Q11
R814 1K_5%
R824 1K_5%
R829 1K_5%
R830 0_5%
AIRACIN
PREP
DOCK_LED_ACT#_R
DOCK_LED_LINK#_R
SPDIF
LINEINL
LINEINR
PR_AOUTL
PR_AOUTR
PDATA(7:0)
SCTS_3R
SRTS_3
SDSR#_3R
RI#_3R
SDCD#_3R
SRXD_3R
STXD_3
SDTR#_3
SLCTIN#_5
PINIT#_5
ERROR#_5
ALF#_5
SLCT_5
PE_5
BUSY_5
ACK#_5
STRB#_5
USB_PN2
USB_PP2
USB_PN3
USB_PP3
PR_HPSENSE#
MBAY_DISABLE
TPB1
TPB1+
TPA1
TPA1+
D+
D-
RD+
RD-
PWR_SWIN#_3
SLP_S3#_5R
PWR_LED_3
KB_DAT_5
KB_CLK_5
EM_DAT_5
EM_CLK_5
SDA_MBAY
SCL_MBAY
COMP_B
CHROMA_C
LUMA_Y
DOCK_VSYNC
DOCK_HSYNC
DOCK_DDCCLK
DOCK_DDCDATA
HPD
VGA_B
VGA_G
VGA_R
DVIDDCCLK_D
DVIDDCDATA_D
TX2M
TX2P
TX1M
TX1P
TXCM
TXCP
TX0M
TX0P
C+
C-
TD+
TD-
VGA_B
VGA_G
VGA_R
DVIDDCCLK
DVIDDCDATA
INVENTEC
MODULES, RUBY1.0
PORT PARALLER
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 61 OF 69
CHANGE by MORRIS CHANG
23-Jul-2003



AMP 20PIN CONNECTOR
MOTHERBOARD SIDE
ACES_88028_2400_24P

INVENTEC			
TITLE: *MODULES, RUBY1.0* *TCPA*			
SIZE: A3	CODE: CS	DOC. NUMBER: *131MBD2600-0-0*	REV: *D*
CHANGE by: *MORRIS CHANG*	23-Jul-2003	SHEET 62 OF 69	



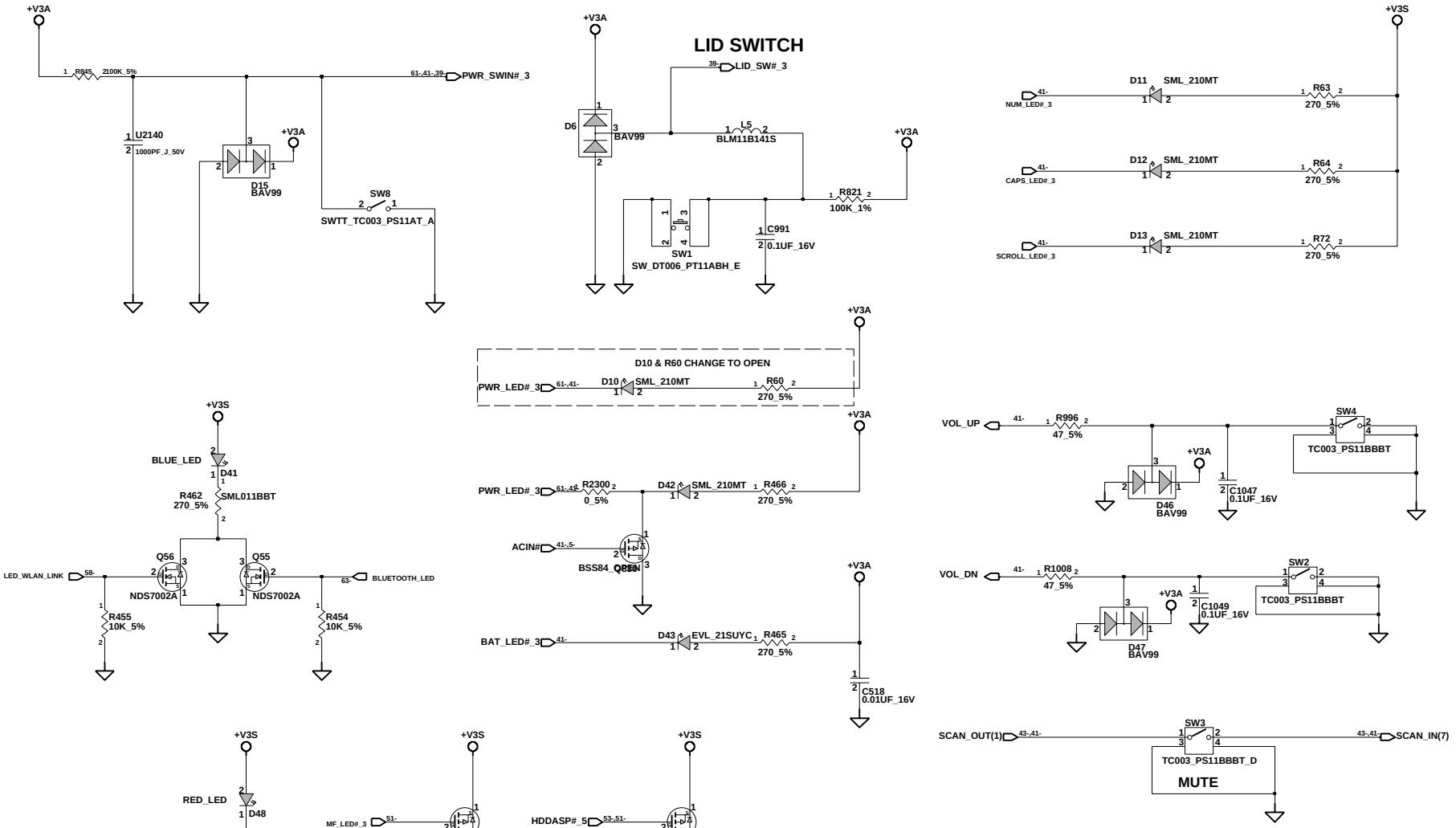
CHANGE by	*MORRIS CHANG*	23-Jul-2003					

INVENTEC

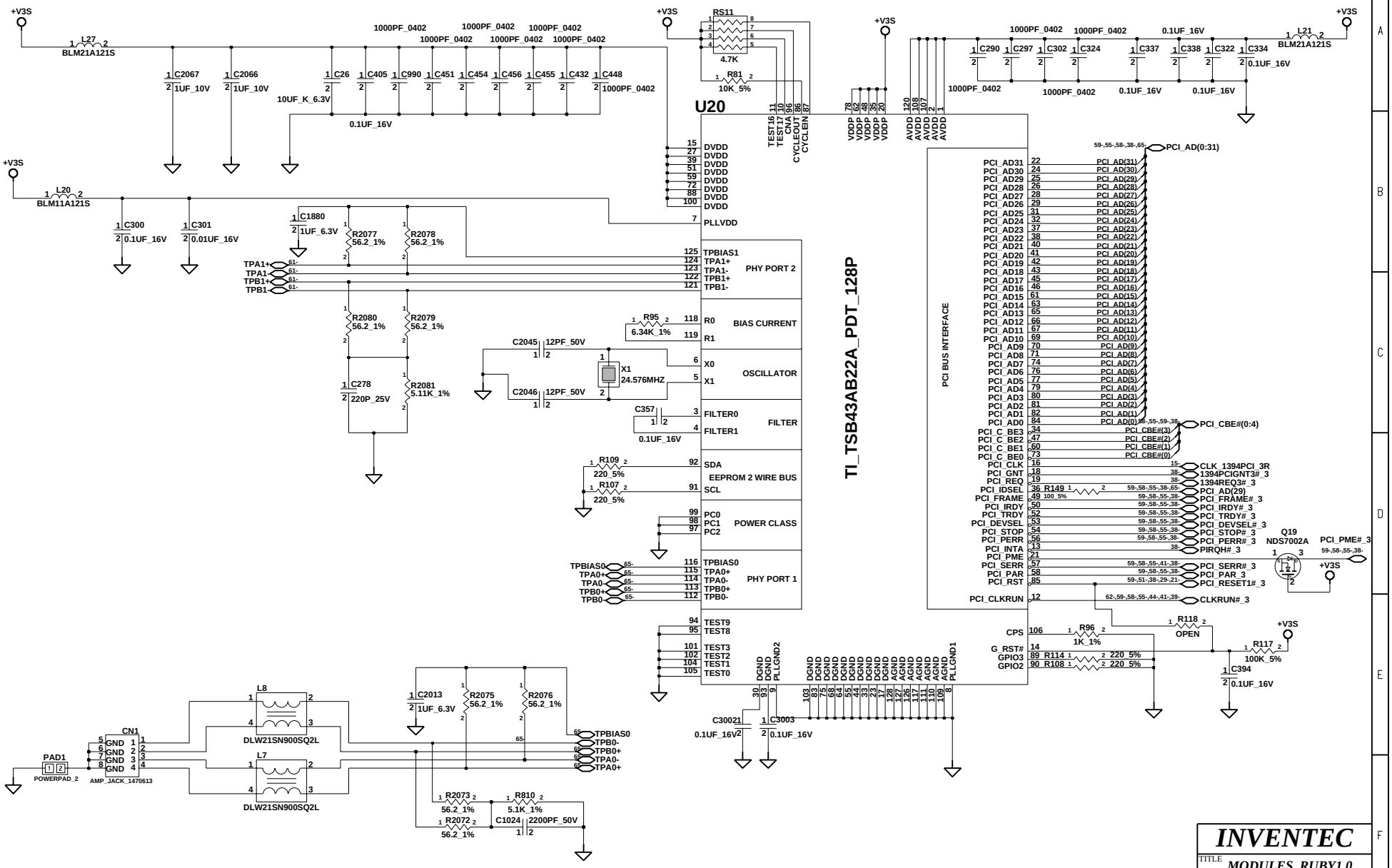
TITLE: ***MODULES, RUBY1.0***
BLUETOOTH

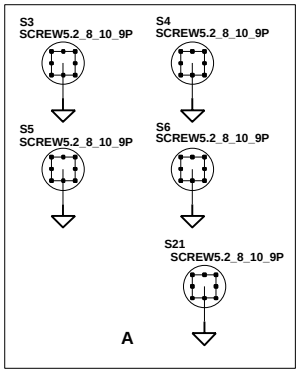
SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

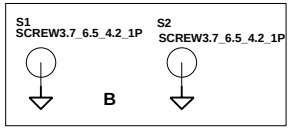
SHEET 63 OF 69

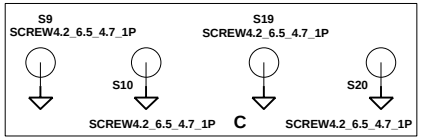


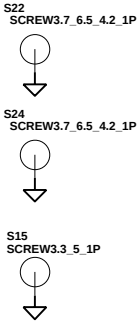
INVENTEC			
TITLE	MODULES, RUBY1.0 SWITCH BUTTON & LED		
SIZE A3	CODE CS	DOC. NUMBER 131MBD2600-0-0	REV D
CHANGE by MORRIS CHANG	23-Jul-2003	SHEET 64 OF 69	

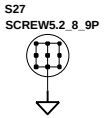

U20
TI_TSB43AB22A_PDT_128P
PHY PORT 2
BIAS CURRENT
OSCILLATOR
FILTER
EEPROM 2 WIRE BUS
POWER CLASS
PHY PORT 1
PCI BUS INTERFACE
PCI_AD(0:31)
PCI_CBE#(0:4)
INVENTEC
MODULES, RUBY1.0
PHY1394 CONTROLLER
SIZE A3
CODE CS
DOC. NUMBER 131MBD2600-0-0
REV D
SHEET 65 OF 69
CHANGE by MORRIS CHANG
23-Jul-2003

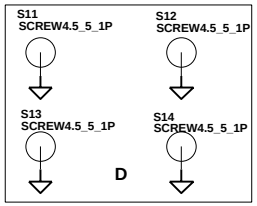


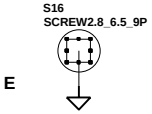


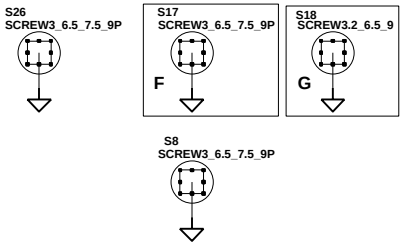




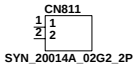




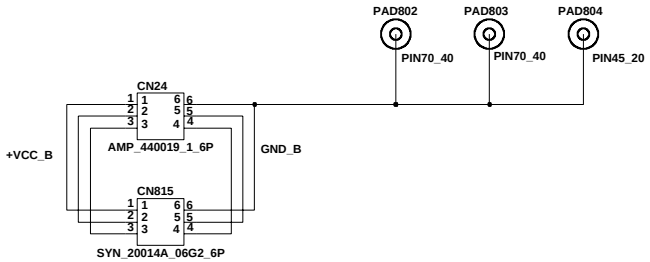




INVENTEC			
TITLE: *MODULES, RUBY1.0* *SCREW*			
SIZE: A3	CODE: CS	DOC. NUMBER: *131MBD2600-0-0*	REV: *D*
CHANGE by: *MORRIS CHANG*	23-Jul-2003	SHEET 66 OF 69	



BATTERY CONNECTOR



6PIN CONNECTOR

CHANGE by	*MORRIS CHANG*	23-Jul-2003				

INVENTEC

TITLE: ***MODULES, RUBY1.0***

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 67 OF 69

SHEET 10

1. R1293 CHANGE FROM 4.99K TO 5.6K
2. R322 CHANGE FROM 10K TO 200K AND ADD CAP(0.01UF) IN R322 PIN2 TO GND

NOTE: FOR ATI POWER SEQUENCE

SHEET 11

1. ADD PM_VGATE DELAY CIRCUIT : U46 , U58 ,R426 , C525 , C526
2. R349 CHANGE FROM 1K TO 0 (OHM) AND ADD CAP (0.01UF) IN R349 RIN2 TO GND

NOTE : FOR PCI_RESET , PCI _CLOCK , PM_VGATE TIMING

SHEET 13

1. R1783 CHANGE FROM 220K TO 100K AND ADD CAP(0.1UF) IN R1783 PIN1 TO GND

NOTE : FOR ATI POWER SEQUENCE(ADJUSTED TIMING OF THE VDD15)

SHEET 14

1. R936 (INPUT TO U40 PIN2) CHANGE FROM 36.5K TO 182(OHM)

NOTE : FOR POWER SEQUENCE

SHEET 15

1. REMOVE Q54 , R390 CHANGE FROM 10K TO 0(OHM)

NOTE :SAME THE DIAMOND SYSTEM

SHEET 17

1. R229 , R230 , R291 CHANGE FROM 0(OHM) TO OPEN

NOTE: MEET THE INTEL RDDP SPEC.

SHEET 20

1. CN8 (FAN CONNECT) CHANGE FROM 5PIN TO 3 PIN

NOTE: MECHANICAL CHANGE

SHEET 29

1. ADD SERIES RESISTORS ON SB_STDS (R518), ADSTBS_O(R519),ADSSTBS_1(R520)
2. CHANGE R894 49.9 TO 40.2(OHM)
3. AGP8X_DETb PULL HIGH POWER RAIL CHANGE FROM +V1.5ATIAGP TO +V3S
4. ADD TEST POINT IN GPIO[4:7] GPIO[9:14]
5. VGA_GPIO[0:3] STATUS CHANGE FROM 10K PULL DOWN TO 10K PULL UP (+V3S)
6. ADD 1K PULL DOWN ON GPIO8
7. ADD 1K PULL DOWN RESISTOR TO PREVENT LEAKAGE FOR SOME ZV_LCDDATA , ZV_LCDCNTL PIN (ZV_LCDDATA[5:15] , [18:19], ZV_LCDCNTL[0:3])
8. ADD 10K PULL DOWN RESISTOR TO PREVENT LEAKAGE FOR ZV_LCDDATA[16:17]

NOTE: ABOVE ITEMS FOR ATI RECOMMED

9. R879 , R880 CHANGE FROM 1K TO 3K

NOTE: FOR +V3A POWER RAIL(ATI-VREFG)

SHEET 30

1. ADD 0 OHM SERIES RESISTOR ON THE PATH OF OSC OUTPUT TO XTALIN AND XTALOUT

NOTE : SELECT FREQUENCY ENTER CHANNEL

2. ADD 10K(R531) PULL UP TO +V3S

NOTE: FOR ATI RECOMMED

SHEET 32

1. L36 CHANGE FROM BLM11A121S TO BLM21A121S

NOTE: FOR ATI RECOMMED

SHEET 33

1. ADD VGA DAMPING PESISTOR

NOTE: FOR ATI RECOMMED

SHEET 39

1. U22-V19(PM_VGATE/VRMPWRGD) CHANGE FROM PM_VGATE TO SB_VGATE
2. (VMEM_CFG3)ADD OPEN(R513) PULL-UP TO +V3S AND 8.2K(R512) PULL-DOWN
3. CREATE RUBY VIDEO MEMORY CALL-BACK DEFINITION TABLE

SHEET 43

1. ROTATE CN16(THE KEYBOARD CONNECT) : 180 POINT OF VIEW
2. ROTATE CN18(THE TOUCH-PAD CONNECT) : 180 POINT OF VIEW
3. CHANGE CN17(STICK POINT) GEOMETRIC(LAYOUT PATTERN)

SHEET 48

1. CHANGE JACK801(MIC JACK) GEOMETRIC(LAYOUT PATTERN)

SHEET 49

1. CHANGE JACK802(EAR-PHONE JACK) GEOMETRIC(LAYOUT PATTERN)

SHEET 51 '53

1. IRQ14_3: CN808(PAGE53) PIN29 CONNECT TO CN814-PIN31(PAGE51)
2. PDCS1#: CN808(PAGE53) PI35 CONNECT TO CN814 -PIN37(PAGE51)
3. HDDASP#_5: CN808(PAGE53) PIN37 CONNECT TO U35-PIN1(PAGE51)
4. PDIAG#: CN808-PIN32(PAGE53) CONNECT TO CN814-PIN34(PAGE51)
5. IDERESET#: R139-PIN1(PAGE53) CONNECT TO U36-PIN4(PAGE51)

NOTE : FOR OPTICAL DRIVE

SHEET 55

1. SWAPPED BELOW SIGNAL : PCI_AD(7)&PCI_AD(8), B_CAD(10)&B_CAD(11), B_CAD(14)&B_CAD(15)

SHEET 65

1. C126 CHANGE VALUE FROM 0.01UF TO 0.1UF

NOTE : REFERENCE TI 1394 CIRCUIT

SHEET 66

1. ADD BELOW SCREW (S21 , S22 , S24 , S26)
2. REMOVE BELOW SCREW (S15)

NOTE: MECHANICAL CHANGE

02 11 CHANGE LIST

SHEET 11

1. R348 TO OPEN

SHEET 29

1. ADD (R521,R522,R523,R524,R525, R527, R528,R529,R530,R534) 0 (OHM) SERIES RESISTOR WOULD BE USEFUL IN TUNING AGP

SHEET 53

1. OPTICAL DRIVE(CN808) CHANGE CONNECT TO AMP_P_012203_50P

SHEET 65

1. U57 ,R526, R535 RESERVE FOR GUID

02 12 CHANGE LIST

SHEET 20

1. CHANGE CN8 PIN2 CONNECTION FROM GND TO Q43 PIN3
   CN8 PIN3 CHANGE TO OPEN

SHEET 39/44

1. ADD INVERTER BETWEEN , PIN H22(GPIO39) OF U22 AND NPCI RESET (PIN26) OF U10. PCI_RESET#_3 WILL NOT LONGER BE CONNECTED TO PIN 26 OF U10

02 14 CHANGE LIST

SHEET 10

1. ADD SCHMITT BUFFER BETWEEN R322 PIN2 AND PIN10(ON1) OF U38

SHEET 26

1. ADD DDR-THERMAL SENSE PARTS(U57, R512,R513)

SHEET 29

1. ADD TEST POINT TO PINS AK19, AH19, AH20,AG20

SHEET 30

1. ADD TEST POINT TO PINS AE25
2. REMOVE C247,C248,R542,R543
3. CHANGE U17 FROM ICS_MK1707S TO SM560

SHEET 39

1. R512 , R513(REFERENCE) CHANGE TO R562 , R563

SHEET 49

1. ADD 0.1UF CAPACITOR BETWEEN ANALOG AND DIGITAL GROUND ON PIN8 OF JACK802

SHEET 53

1. ADD 1K RESISTOR PULL-DOWN TO PIN47 OF CN808

SHEET 65

1. ADD 1K RESISTOR PULL-DOWN TO PIN7 OF U57
2. CN1 PINOUTS ARE REVERSED
3. L9(BLM11A121S) CHANGE TO POWER_PAD2
4. U57(REFERENCE) CHANGE TO U61

SHEET 66

1. ADD SCREW (S15))

SHEET 39/53

1. U22 PIN G23(GPIO40) CONNECT TO CN808 PIN47

02 17 CHANGE LIST

SHEET 53

1. R561 1K CHANGE TO 100K

SHEET 68

1. CN15 PIN12 CONNECT TO GND

CHANGE by	*MORRIS CHANG*	23-Jul-2003

INVENTEC

TITLE: *MODULES, RUBY1.0*
CHANGE LIST

SIZE	CODE	DOC. NUMBER	REV
A3	CS	*131MBD2600-0-0*	*D*

SHEET 68 OF 69

RUBY SI BUILD CHANGE LIST

1. ADD OPC CIRCUIT ON PAGE5.
2. R2169 CHANGE FROM 10K TO OPEN ---PAGE6
3. REMOVE D22 ---PAGE6
4. U28-PIN8 CONNECTION STATUS : CHANGE IT FROM R2169-PIN1 TO Q802-PIN1.2.3 ---PAGE6
5. Q22-PIN1.2.3 AND Q26-PIN1.2.3 POWER NAME, CHANGE IT FORM +VBATR TO +VADP2 ---PAGE6
6. REMOVE PAD9 ---PAGE11
7. REMOVE PAD11 AND PAD12 ---PAGE12
8. CHANGE MOSFET TO REDUCE RDS ON Q809 Q39 ---PAGE13
9. CHANGE THERMAL IC PART, FROM ADM1032 TO ADM 1031 FOR CPU&VAG ---PAGE20
10. ADD TEST-POINT (TP557) ---PAGE21
11. ADD CAP (220PF) 3 PCS BETWEEN THE MCH_GTLREF AND GND ---PAGE22
12. REMOVE C60 (150UF) ---PAGE26
13. DELETE THERMAL SENSOR CIRCUIT ---PAGE30
14. R468~R489, R490~R492, R494~R511 : CHANGE THESE RESISTOR FROM 0(OHM) TO 10(OHM) ---PAGE33
15. ADD CAP (150UF)BETWEEN THE +VRAM_VCC AND GND ---PAGE34
16. PIN1 OF D2,D3,D800,D801,D802,D803,D804 ARE CHANGE FROM +V3 TO USBCC1 ---PAGE36
17. A_CSTOP# , B_CSTOP# OF U2228 SIGNAL(PAGE55) CONNECT TO CN21(PAGE56)
18. CHANGE CN10 SYMBOL : PIN11 IS REF_P_D AND PIN12 IS WP ---PAGE57
19. REMOVE C230, C395, C1674, C1675. CHAISY _GND CHANGE TO DIGITAL_GND ---PAGE60
20. UPDATE CN807 FROM JAE_WD_154S4V_VF78_ML_154P TO JAE_WD_154S4VH_VF85_N_154P AND ADD D+, D-, C+, C- SIGNALS FROM U2(PAGE60) ---PAGE61
21. BREAKING PIN1 AND PIN3 CONNECTION FOR SW2, SW3, SW4 THEN ALL OF PIN3 CONNECT TO GND ---PAGE64
22. UPDATE CN1 SYMBOL, ADD GND SIGNAL OF PIN7, PIN8 THEN CONNECT TO GND ---PAGE65
23. ADD NAME PORT(MAX_LX5) IN L44-PIN1 ...PAGE7
24. REMOVE NONE-USE SIGNAL-PORT(SLP_S5#_5R) ...PAGE13
25. Q48 CHANGE FROM FDC638P TO FDR840P ...PAGE13
26. R293 CHANGE FROM 10K TO OPEN ...PAGE17
27. Q806 CHANGE FROM FDC638P TO FDR840P(Q825) ...PAGE37
28. R848-PIN1 AND Q825-PIN5,PIN8 POWER RAIL FROM +V3S TO +V3 ...PAGE37
29. R67 CHANGE VALUE FROM 180K TO 51K ...PAGE39
30. REMOVE D39 ...PAGE41
31. ADD DELAY CIRCUIT FOR THE PWR_GOOD3 ...PAGE41
32. ADD RNET FOR KAHUNA PROGRAMMING(RS850,RS860) ...PAGE43
33. REMOVE D17 ...PAGE52
34. U7 AND U804 PIN5,PIN7 POWER RAIL CHANGE FROM +V5 TO +V5A ; OC# SIGNAL OF THE PIN2 IS REMOVE EN SIGNAL OF THE PIN1 IS CONNECT TO +V5 ...PAGE54
35. REMOVE R934 R946 OF THE PULL-HIGH CIRCUIT ...PAGE56
36. CHANGE TURN ON CIRCUIT OF THE H.D.D-LED, USING A MOS-FET(Q286) TO CONTROL H.D.D-LED ...PAGE64
37. R1679 CHANGE VALUE FROM 20K TO 1.3K ; R2139 CHANGE VALUE FROM 150K TO 10K ...PAGE5
38. DELETE R514 ...PAGE26
39. FOLLOWIG RESISTORS ARE CHANGE VALUE FROM 10 OHM TO 24 OHM : RS12~RS20 , RS22~RS24 ; RS27~RS30 RS27~RS30 ; RS32~RS35 ; RS38~RS44 ; RS46~RS47 ; RS57~RS59 ; R481~R486 ; R488 , R489 ; R505~R511 ; R502 ...PAGE33
40. ADD CAP(22UF) BETWEEN THE +V2.5S AND GND IN U23 , U27 OF THE VRAM ...PAGE34
41. ADD CAP(22UF) BETWWEEN THE +V2.5S AND GND IN U14 , U15 OF THE VRAM ...PAGE35
42. CHANGE CRT CIRCUIT, REFER TO THE DIAMOND CRT CIRCUIT ...PAGE36
43. R67 CHANGE VALUE 51K TO 180K ...PAGE39
44. POWER RAIL OF THE R223-PIN1, CHANGE VALUE FROM +V5A TO +V5 ...PAGE46
45. POWER RAIL OF THE U32-PIN5, CHANGE VALUE FROM +V5 TO +V3 ...PAGE57
46. C1620 , C1621 , C1623 CHANGE VALUE FROM 0.01UF TO 0.1UF ...PAGE59
47. C1651 , C1656 CHANGE VALUE FROM 22PF TO 27PF ...PAGE60
48. ADD 200 OHM SERIES RESISTOR BETWEEN XTALO AND PIN2 OF X800(CRYSTAL) ...PAGE60
49. D41 CHANGE FROM SML_210MT TO SML010BAT(BLUE-LED) ; D48 CHANGE FROM SML_210MT TO 12-215VRC_TR8(RED-LED) ...PAGE64
50. DELETE U8 , R526 , R535 , R538 ...PAGE65
51. CN1 CHANGE FROM AMP_JACK_440148 TO AMP_JACK_1470613 ...PAGE65
52. CN4 CHANGE FROM MLX_53398_0490 TO JST_BM4B_SRSS ...PAGE50

CHANGE by	*MORRIS CHANG*	23-Jul-2003	

INVENTEC

TITLE	*MODULES, RUBY1.0 CHANGE LIST*		
SIZE A3	CODE CS	DOC. NUMBER *131MBD2600-0-0*	REV *D*
		SHEET 69 OF 69	